



Who is ready for CBDC? A multi-pillar cross-country readiness index and toolkit for policy makers

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Abstract

This paper introduces a comprehensive, multi-pillar Central Bank Digital Currency (CBDC) Readiness Index, designed to assess the foundational capacity of countries for implementing CBDCs. The index comprises five pillars: Digital and Technology Infrastructure, Economic and Household Capacity, Financial Depth and Inclusion, Institutions and Regulation, and Macro Resilience and Policy, providing a data-driven framework for evaluating CBDC readiness. Unlike existing indices that focus on project status or attention measures, this index emphasizes long-term structural factors such as digital infrastructure, financial inclusion, and institutional strength. Constructed using a robust, auditable methodology, the index employs CRITIC information weighting and geometric aggregation to ensure transparency and reliability. Calculated for up to 183 economies, the index allows for a comprehensive assessment of CBDC readiness, distinguishing it from individual design choices or short-term announcements. Robustness and sensitivity checks, including bootstrap resampling and leave-one-variable-out experiments, validate the stability of rankings. External validation against an independent dataset of CBDC projects, search interest and central-bank communication confirms that the index is positively associated with observed CBDC activity while remaining distinct from generic economic development, and a cluster analysis groups economies into four readiness archetypes that translate the index into differentiated policy guidance. The CBDC Readiness Index serves as a policy toolkit, enabling central banks and international institutions to identify digital, financial, and institutional barriers and to design CBDC projects aligned with country-specific readiness profiles.

Keywords CBDC readiness · Policy design · Multi-pillar index · Central banks · Financial stability

JEL Classification E42 · E58 · G20 · O33

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1 Introduction

Central banks are moving from debating central bank digital currencies (CBDCs) to designing and piloting them. BIS surveys document a rapid transition from exploration to advanced experimentation and live pilots, with motivations broadening from defensive reactions to crypto-assets toward structural goals such as payment resilience, cross-border efficiency, inclusion and monetary sovereignty (Auer et al. 2021; Boar et al. 2020; Illes et al. 2024; Kosse and Mattei 2022). IMF and central-bank research stresses that CBDC design is an instrument for delivering policy objectives rather than a purely technological choice (Agur et al. 2019; Mancini-Griffoli et al. 2018; Tan 2024).

The theoretical literature formalises these questions across macroeconomic, banking and institutional frameworks. DSGE and open-economy models analyse effects on monetary transmission, money demand, currency substitution and capital flows (Barrdear and Kumhof 2022; Chen and Siklos 2022; Karau 2023; Mishra and Prasad 2024; Xiang et al. 2024), and surveys emphasise that CBDC implications depend on design choices such as access, remuneration and holding limits, with tiered remuneration and two-tier models proposed to mitigate disintermediation (Auer et al. 2022; Bindseil 2020; Chiu et al. 2023; Niepelt 2020; Williamson 2022). CBDC is thus a structural intervention in the liability composition of the financial system whose consequences depend on both design and institutional conditions.

A growing literature shows that CBDC introduction shifts the competitive environment of banking intermediation: a direct central bank liability can create funding-substitution pressures that alter deposit pricing, liquidity management and credit supply (Andolfatto 2021; Keister and Monnet 2022; Kim and Kwon 2023; Nyffenegger 2024). The capacity of banks to absorb this change becomes a determinant of systemic resilience (Palmieri and Geretto 2023), so readiness cannot be reduced to technological availability or financial size.

Empirical evidence from pilots and cross-country panels supports this state-dependent view: CBDC initiatives can strengthen stability and credit allocation in some contexts while generating funding pressures in others (Luu et al. 2023; Meng et al. 2024; Xiao et al. 2024), and CBDC-related announcements and uncertainty act as risk factors for systemic risk and asset markets (Bas et al. 2024; Hoque et al. 2024; Wang et al. 2022). These heterogeneous outcomes reinforce that CBDC effects are conditional on underlying structural fundamentals.

A complementary strand examines adoption, trust and privacy. Usage intentions depend on perceived usefulness, security and institutional credibility, and demand for central bank liabilities hinges on perceptions of safety and privacy relative to private alternatives (Infante et al. 2022; Chua et al. 2025; Liu et al. 2024). This work implies that formal frameworks alone are insufficient: institutional legitimacy and public confidence are integral to CBDC viability.

Despite this rich theoretical landscape, empirical measurement of CBDC preparedness remains fragmented. BIS surveys map project stages (Boar et al. 2020; Illes et al. 2024; Kosse and Mattei 2022; Auer et al. 2021), news-based indices capture attention and uncertainty (Bas et al. 2024; Wang et al. 2022), and country-characteristics studies identify correlates of adoption (Alfar et al. 2023; Tong and Jiayou 2021). Policymakers still lack a systematic cross-country diagnostic of the structural preconditions that condition the feasibility and stability of CBDC introduction.

This paper develops a multi-pillar CBDC Readiness Index that operationalizes structural preparedness across five domains: Digital and Technology Infrastructure; Economic and Household Capacity; Financial Depth and Inclusion; Institutions and Regulation; and Macro Resilience and Policy. Readiness in this framework is defined as the presence of foundational structural conditions that enable a jurisdiction to introduce and operate a CBDC within its monetary and financial system without generating destabilizing frictions. The index deliberately focuses on baseline structural capacity rather than on specific design choices, remuneration policies or short-term political decisions. It measures core readiness, understood as the necessary but not sufficient condition for safe and effective CBDC implementation.

The selection of these five domains follows a necessity logic grounded in the operation of a CBDC rather than in general economic performance. A jurisdiction must process digital payments securely at scale (Digital and Technology Infrastructure), its population must access and use digital money (Economic and Household Capacity), its financial system must be deep and inclusive enough to absorb shifts in liability composition without disorderly disintermediation (Financial Depth and Inclusion), its legal framework must confer issuance authority, consumer protection and dispute resolution (Institutions and Regulation), and its macroeconomic environment must be stable enough that a new instrument does not amplify currency or deposit substitution (Macro Resilience and Policy). Each domain is necessary; none is individually sufficient, and the five are not asserted to be jointly sufficient, since design choices, political will and timing lie outside structural capacity. Design-stage dimensions such as the remuneration rule, token versus account architecture, or programmability are excluded by construction as decisions taken once a jurisdiction is structurally ready.

This necessity logic also clarifies the apparent counterexamples in which advanced economies remain cautious while some structurally weaker economies launch early. Readiness measures whether the foundational conditions are present, not whether a jurisdiction has chosen to act on them. A highly ready economy may defer issuance for reasons of monetary-policy caution, financial-stability conservatism or the adequacy of existing fast-payment rails, while a less ready economy may launch in response to financial-inclusion or monetary-sovereignty pressures despite incomplete foundations. Observed launch behaviour is thus an imperfect signal of readiness, and the gap between the two is precisely what a structural index is designed to expose. The external validation reported in Sect. 4.6 confirms this interpretation empirically, showing that readiness is positively but only moderately associated with observed project advancement and is not a restatement of development.

The index does not measure behavioural demand or central-bank design sophistication, which are endogenous to policy choices and may evolve after structural readiness is established. It isolates the foundational conditions that precede and condition higher-order instruments such as tiered remuneration and liquidity backstops.

Our contributions are threefold. First, we provide a conceptually explicit and empirically tractable definition of structural CBDC readiness that bridges monetary theory, banking intermediation and institutional economics. Second, we construct a cross-country index using a transparent and auditable methodology combining robust normalization, cost-direction alignment, CRITIC information weighting and non-compensatory geometric aggregation. Third, we conduct extensive robustness analyses including alternative weighting schemes, alternative aggregation rules, leave-one-variable and leave-one-pillar-out experiments, bootstrap rank stability and measurement perturbation diagnostics. These exercises

confirm that leader and laggard classifications are highly robust while intermediate rankings remain appropriately sensitive to specification choices.

Fourth, we externally validate the index against an independent dataset of CBDC projects, public search interest and central-bank communication, and we show through partial-correlation and nested-regression evidence that readiness is associated with observed CBDC activity beyond what economic development alone explains. We complement the ranking with a cluster analysis that organizes economies into four readiness archetypes, providing a typology that is more directly actionable for policy sequencing than ordinal position.

The index is intended as a structural diagnostic dashboard rather than a predictive model of adoption timing. It complements macroeconomic simulations, banking studies and sentiment-based indices by benchmarking the underlying conditions that shape heterogeneous outcomes (Bas et al. 2024; Luu et al. 2023; Wang et al. 2022).

The remainder of the paper proceeds as follows. Section 2 reviews the CBDC policy, macro-financial and adoption literatures. Section 3 describes data and methodology. Section 4 presents baseline results. Section 5 reports robustness analyses. Section 6 discusses policy implications and concludes, including limitations regarding behavioural demand, political legitimacy and dynamic adaptation processes that lie beyond the structural focus of this study.

2 Literature review

Central banks' CBDC research now sits at the intersection of monetary theory, financial intermediation, digital infrastructure, and political economy. The literature has evolved from exploratory policy discussion toward formal modelling of monetary transmission, banking adaptation, macro-financial spillovers, and behavioural adoption. In this section, we position the CBDC Readiness Index within four interrelated strands: (i) policy and design frameworks from IMF, BIS and central banking research; (ii) banking intermediation, adaptation and macro-financial transmission; (iii) adoption, privacy and institutional trust; and (iv) CBDC-related indices and cross-country comparisons. We then clarify how a structural, multi-pillar readiness index complements and extends these approaches.

2.1 Policy motives, design architecture and institutional framing

IMF and BIS research has shaped the foundational policy narrative around CBDC. IMF contributions emphasise that CBDC design must be evaluated relative to public policy objectives including financial inclusion, efficiency, competition and stability rather than as a purely technological innovation (Agur et al. 2019; Mancini-Griffoli et al. 2018; Tan 2024). BIS surveys document the transition from exploratory investigations to pilot implementation, with motivations shifting from defensive responses to crypto-assets toward structural goals such as payment resilience and cross-border efficiency (Auer et al. 2021; Boar et al. 2020; Illes et al. 2024; Kosse and Mattei 2022).

Academic work translates these motivations into formal frameworks. Auer et al. (2022) show that the macroeconomic implications of CBDC are architecture dependent, since a direct or two-tier model alters transmission, liquidity flows and bank funding. DSGE and open-economy models examine monetary transmission, money demand, currency substi-

tution and capital flows (Barrdear and Kumhof 2022; Chen and Siklos 2022; Mishra and Prasad 2024; Xiang et al. 2024), while design-focused studies analyse remuneration, holding limits and tiered structures (Andolfatto 2021; Bindseil 2020; Chiu et al. 2023; Niepelt 2020; Williamson 2022).

In this literature, macroeconomic conditions are not exogenous determinants of CBDC success but structural context variables that condition the credibility of policy calibration: inflation stability proxies monetary credibility, and external balance reflects vulnerability to currency substitution and cross-border dynamics (Auer et al. 2022; Mishra and Prasad 2024).

Legal scholarship further highlights mandate constraints, liability design and constitutional compatibility (Bossu et al. 2020; Cullen 2022; Nabilou 2020). Although this literature rigorously analyses design trade-offs, it typically treats digital infrastructure, financial depth and institutional quality as stylised background parameters rather than as systematically measured cross-country fundamentals.

2.2 Banking intermediation, adaptation and macro-financial stability

A second strand examines CBDC through banking intermediation and financial stability. Theoretical models study disintermediation risk when CBDC competes with deposits and whether two-tier architectures and calibrated remuneration preserve credit supply (Andolfatto 2021; Chiu et al. 2023; Keister and Monnet 2022; Kim and Kwon 2023; Nyffenegger 2024), and open-economy models explore cross-border spillovers in multi-CBDC settings (Chen and Siklos 2023; Karau 2023; Moro and Nispi Landi 2024). These models underscore that CBDC alters the competitive environment for banks, with deposit funding and liquidity management adjusting accordingly (Palmieri and Geretto 2023).

From a measurement perspective, the Financial Depth and Inclusion pillar captures baseline intermediation capacity and retail access through broad money, private credit, branch density and payment-usage indicators. These are necessary conditions for CBDC circulation corridors but do not capture banks' strategic adaptability, which is a partly endogenous, post-introduction process treated as a limitation.

Empirically, cross-country panels suggest CBDC adoption can enhance stability in some contexts, particularly for diversified banks and emerging markets (Luu et al. 2023), and e-CNY analyses document improved credit allocation (Meng et al. 2024; Xiao et al. 2024). At the macro-financial frontier, CBDC announcements and uncertainty act as risk factors for systemic risk and asset markets (Bas et al. 2024; Hoque et al. 2024; Wang et al. 2022). These outcomes are conditional on both design calibration and structural fundamentals.

2.3 Adoption, institutional trust and the privacy paradox

A third strand addresses households and firms. Technology-acceptance research identifies perceived usefulness, security and ease of use as dominant adoption determinants (Gupta and Pandey 2023; Liu et al. 2025; Wang et al. 2023a, b), with trust formation depending on both individual disposition and institutional credibility (Chua et al. 2025; Osakwe et al. 2025).

Demand for central bank liabilities depends on perceptions of safety and privacy relative to private alternatives (Infante et al. 2022), and legal analyses document tensions between

AML and CFT compliance and transactional anonymity (Pocher and Veneris 2022; Soana and de Arruda 2024).

These findings distinguish implementation capacity from behavioural acceptance: a jurisdiction may possess advanced infrastructure and formal frameworks yet face scepticism over surveillance or data governance. The Institutions and Regulation pillar captures formal governance quality and regulatory clarity, which are structural prerequisites for credible deployment, not forecasts of retail demand (Infante et al. 2022).

2.4 CBDC indices, attention measures and cross-country comparisons

Recent work increasingly relies on aggregate CBDC indices: BIS surveys document project stages (Boar et al. 2020; Illes et al. 2024; Kosse and Mattei 2022), news-based indices measure attention and uncertainty spillovers (Bas et al. 2024; Wang et al. 2022), and adoption indices examine bank stability and inclusion (Luu et al. 2023; Dunbar and Treku 2024).

Cross-country studies identify correlates of advanced CBDC stages (Alfar et al. 2023; Tong and Jiayou 2021) and situate CBDC within currency hierarchies (Chen and Siklos 2023; Kuehnlenz et al. 2023), while bibliometric surveys emphasise the absence of comprehensive structural indicators for benchmarking preparedness (Bhaskar et al. 2022; Dionysopoulos et al. 2024; Kumar and Rani 2025).

2.5 Contribution of a structural multi-pillar readiness framework

The readiness index contributes along three dimensions. First, it operationalises structural preparedness rather than project activity, integrating digital infrastructure, household capacity, financial depth, regulation and macro resilience into a high-dimensional benchmark. Second, it complements design-focused theory, since the effectiveness of tiered remuneration and liquidity backstops depends on baseline institutional credibility, financial integration and macro stability (Bindseil 2020; Chiu et al. 2023). Third, it clarifies scope: it measures core structural readiness and does not quantify behavioural demand, political legitimacy or bank-level adaptation, which are endogenous and addressed as limitations.

In this sense, the CBDC Readiness Index functions as a structural diagnostic dashboard. It complements macroeconomic simulations, banking studies and attention-based indices by anchoring them in measurable cross-country fundamentals. By distinguishing structural capacity from design calibration and behavioural acceptance, it provides policymakers with a transparent and robust baseline assessment of preparedness.

3 Methods

We construct a multi-pillar CBDC Readiness Index using a cross-country panel, employing a fully reproducible and auditable pipeline. The exposition is organised into three focused subsections: (i) samples and data mapping; (ii) per-year preprocessing and normalization; and (iii) index construction together with a comprehensive robustness and uncertainty programme.

Table 1 Pillar-to-variable conceptual mapping (theoretical channel and normalisation direction). Variable definitions are listed in the table below; descriptive statistics are reported separately in Table 1B in Sect. 4.1

Pillar	Variable	Short form	Scale / Units	What it means
Digital & Tech Infrastructure	Fixed broadband subscriptions (per 100 people)	Broadband_Subscriptions	per 100 people	Penetration of fixed broadband among the population.
	Individuals using the Internet (% of population)	Internet_Use	percent (%)	Share of people who used the Internet in the past 3 months.
	Mobile cellular subscriptions (per 100 people)	Mobile_Subscriptions	per 100 people	Number of active mobile-cellular subscriptions relative to population.
	Secure Internet servers (per 1 million people)	Secure_Servers	per 1,000,000 people	Density of TLS/SSL servers; proxy for secure online ecosystem.
	annual_foreign_investment_telecom	ForeignInv_Telecom	US\$ (current), annual flow	Foreign investment directed to telecom sector.
	international_bandwidth_usage	Intl_Bandwidth	Mbps/Gbps (level)	International Internet bandwidth in use; external connectivity capacity.
	mobile_bb_active_per100	Mobile_BB_Active	per 100 people	Active mobile-broadband subscriptions per 100 inhabitants.
	p3_smartphone_ownership_pct	smartphone	percent (%)	Share of individuals who own a smartphone.
Economic & Household Capacity	Access to electricity (% of population)	Access_Electricity	percent (%)	Population with access to electricity.
	GDP per capita (current US)	GDP_per_Capita	Average income per person in current dollars.	
	GDP per capita growth (annual %)	GDP_perCapita_Growth	percent (%)	Annual growth rate of real GDP per capita.
	Gini index	Income_Inequality	index (0–100)	Income inequality (higher = more unequal). Treated as cost in normalization.
	Urban population (% of total population)	Urban_Population	percent (%)	Share of population living in urban areas.
Financial Depth & Inclusion	Account ownership at a financial institution or with a mobile-money-service provider (% age 15+)	Account_Ownership	percent (%)	Adults with an account (bank or mobile money).
	Broad money (% of GDP)	Broad_Money	percent of GDP	Size of broad money relative to economy; proxy for financial depth.
	Commercial bank branches (per 100,000 adults)	Bank_Branches	per 100,000 adults	Density of brick-and-mortar banking access points.
	Domestic credit to private sector (% of GDP)	Private_Credit	percent of GDP	Credit provided to private sector by financial institutions.

Table 1 (continued)

	Made or received a digital payment (% age	Digital_Payments	percent (%)	Adults who made/received any digital payment.
	Digitally enabled account (% age 15+)	Digital_Accounts	percent (%)	Adults with an account that can be used digitally.
	Made a digital merchant payment (% age 15+)	Merchant_Payments	percent (%)	Adults who paid a merchant digitally.
	Received government payments: into an account (% age 15+)	Govt_Payments_Account	percent (%)	Adults receiving government transfers directly to an account.
	Received government payments: in cash only (% age 15+)	Govt_Payments_Cash	percent (%)	Adults receiving government transfers only in cash (cost; inverted).
Institutions & Regulation	Political Stability and Absence of Violence/Terrorism: Estimate	Political_Stability	WGI estimate (≈ -2.5 to $+2.5$)	Perception of political stability (higher is better).
	Regulatory Quality: Estimate	Regulatory_Quality	WGI estimate (≈ -2.5 to $+2.5$)	Perception of regulatory quality (higher is better).
	Rule of Law: Estimate	Rule_of_Law	WGI estimate (≈ -2.5 to $+2.5$)	Perception of rule of law (higher is better).
	consumer_protection_framework_flag	Consumer_Protection	binary (0/1)	Legal consumer protection framework exists (1) or not (0).
	cybersecurity_framework_flag	Cybersecurity_Framework	binary (0/1)	National cybersecurity strategy/framework exists (1) or not (0).
	data_policies_flag	Data_Policies	binary (0/1)	Data governance/privacy policies exist (1) or not (0).
	digital_dev_strategy_flag	Digital_Dev_Strategy	binary (0/1)	National digital development strategy exists (1) or not (0).
	digital_id_framework_flag	Digital_ID_Framework	binary (0/1)	National digital ID framework exists (1) or not (0).
	dispute_resolution_mechanisms_flag	Dispute_Resolution	binary (0/1)	Regulatory dispute resolution mechanism exists (1) or not (0).
	eapplications_framework_flag	EApps_Framework	binary (0/1)	Regulatory framework for e-applications exists (1) or not (0).
	enforcement_appeals_flag	Enforcement_Appeals	binary (0/1)	Enforcement and appeals mechanism exists (1) or not (0).
	foreign_participation_flag	Foreign_Participation	binary (0/1)	Foreign participation/investment permitted (1) or not (0).
	ict_policies_flag	ICT_Policies	binary (0/1)	National ICT policy framework exists (1) or not (0).
	infra_sharing_framework_flag	Infra_Sharing	binary (0/1)	Infrastructure sharing regulation exists (1) or not (0).
	internet_governance_framework_flag	Internet_Governance	binary (0/1)	Internet governance framework exists (1) or not (0).

Table 1 (continued)

	intl_connectivity_access_flag	Intl_Connectivity_Access	binary (0/1)	Mandate ensuring access to international connectivity exists (1) or not (0).
	intl_facilities_access_mandate_flag	Intl_Facilities_Access	binary (0/1)	Mandated access to international facilities exists (1) or not (0).
	qos_regulatory_framework_flag	QoS_Regulation	binary (0/1)	Quality-of-Service regulation exists (1) or not (0).
	regulatory_roadmaps_flag	Regulatory_Roadmap	binary (0/1)	Published regulatory roadmap exists (1) or not (0).
Macro Resilience & Policy	External balance on goods and services (% of GDP)	External_Balance	percent of GDP	Net exports of goods and services relative to GDP.
	Inflation, consumer prices (annual %)	Inflation	percent (%)	Annual CPI inflation (cost; inverted).
	Real effective exchange rate index (2010 = 100)	REER_Index	index (2010=100)	REER level; deviation from long-run reference penalised in either direction (Cost on deviation).

Notes - *Binary policy flags* are included as-is (0/1). If a flag shows no variation in a given year (all 1s or all 0s), it carries zero information for CRITIC weighting that year. - *Cost indicators* (e.g., inflation, cash-only transfers) are inverted during normalization so that higher standardized values always signal **greater CBDC readiness**.

3.1 Data and samples

The analysis uses a full master panel (FULL_183) and three coverage-restricted subsamples defined by per-country coverage thresholds of about 40, 50 and 70 percent (SAMPLE_40, SAMPLE_50, SAMPLE_70); the countries in each sample are listed in Appendix A. A BIS speeches subset isolates country-years in which central-bank speeches explicitly mention CBDC and is used for a targeted robustness exercise, detailed in Appendix B. The mapping from raw inputs to the five pillars is presented in Table 1.

These thresholds correspond to quartile-aligned breakpoints in the distribution of indicator coverage: the 40 percent cut isolates structurally sparse economies (mainly small island and low-income states), the 50 percent cut sits near the median, and the 70 percent cut identifies the high-coverage cluster of OECD members and large emerging markets. Reporting all three lets the reader observe how the index behaves across the coverage distribution.

Pillars and variable selection. The index is structured around five pillars that reflect how central banks, international institutions and research frame the preconditions for CBDC deployment: Digital and Technology Infrastructure; Economic and Household Capacity; Financial Depth and Inclusion; Institutions and Regulation; and Macro Resilience and Policy. Surveys and bibliometric reviews show CBDC debates cluster around these domains rather than isolated indicators (Bhaskar et al. 2022; Dionysopoulos et al. 2024; Kumar and

Rani 2025), and BIS and IMF frameworks emphasise the same structural preconditions (Das et al. 2023). We map these strands into observable indicators in Table 2.

Variable selection follows four principles. First, each indicator must have a clear conceptual channel to CBDC design, adoption or transmission (Barrdear and Kumhof 2022; Chiu et al. 2023; Wang et al. 2023a, b). Second, indicators must be consistently measurable across a large multi-year panel (Alfar et al. 2023; Nanez Alonso et al. 2021). Third, mechanically design-specific variables such as remuneration rates or holding limits are excluded, since the index measures baseline readiness rather than design quality. Fourth, because the index uses CRITIC weighting and geometric aggregation, we select a parsimonious but information-rich set within each pillar.

Variable selection is theory-driven while weighting is data-driven: the conceptual filter admits only indicators with a defensible channel, and CRITIC then sets weights from observed dispersion and pairwise correlation. This two-stage separation keeps weights auditable each year. The trade-off is that a theoretically important indicator can carry a weight that depends on that year's cross-country dispersion; we retain CRITIC for reproducibility and benchmark a theory-weighted variant in Sect. 4.3.

Digital and Technology Infrastructure. This pillar captures the operational foundation for scalable, secure and inclusive CBDC deployment. Broadband penetration, mobile connectivity, server density and international bandwidth are preconditions for robust functionality and integration with existing payment ecosystems (Li et al. 2022; Peng et al. 2025; Wu et al. 2024). Smartphone ownership and active mobile broadband matter because most pilots use wallet-based mobile front ends (Lee et al. 2021), secure servers proxy cybersecurity maturity (Auer et al. 2021), and foreign telecom investment and international bandwidth capture external connectivity supporting cross-border arrangements (Kuehnlenz et al. 2023). The eight indicators measure infrastructure capacity rather than institutional trust: whether a jurisdiction has the hardware and connectivity to operate a CBDC securely and at scale, not whether technology alone guarantees acceptance or stability.

Economic and Household Capacity. CBDC adoption depends on household and firm-level absorptive capacity. Access to electricity is a basic prerequisite for digital payments (Agur et al. 2019), while GDP per capita and its growth proxy fiscal capacity, technological absorption and demand for advanced financial services, all associated with progression to advanced CBDC stages (Alfar et al. 2023). Income inequality, entered as a cost indicator and inverted in normalisation, reflects structural frictions in inclusion, and urban population shares proxy agglomeration and network effects. The pillar captures absorptive capacity and distributional context rather than behavioural demand.

Financial Depth and Inclusion. This pillar connects CBDC to banking intermediation and payment provision. CBDC introduction can affect deposit rates, funding structures and lending, with the direction depending on remuneration design, holding limits and pre-existing depth (Andolfatto 2021; Barrdear and Kumhof 2022; Chiu et al. 2023; Nyffenegger 2024); disintermediation risk can be mitigated through tiered remuneration but depends on the resilience of the banking system (Bindseil 2020). The pillar measures four observable layers: macro intermediation scale (broad money, private credit), physical access (bank branches), account-level inclusion (account ownership, digital accounts), and transactional digitalisation (digital and merchant payments, government-transfer modality). These proxy structural absorptive capacity for CBDC corridors rather than the post-introduction adaptability of individual banks.

Table 2 Descriptive statistics for input indicators in the full sample (183 countries). Separated from the conceptual mapping in Table 1. Variables with notably uneven coverage are visible in the count column

Variable	Count	Mean	Median	Std	Min	Max
Broadband_Subscriptions	908	16.51255	12.9182	15.32107	0	55.8836
Internet_Use	936	66.5561	74.9843	26.91878	4	100
Mobile_Subscriptions	917	110.2784	114.677	33.23586	18.9298	214.842
Secure_Servers	1104	26,922.03	625.3288	142,508.1	0.433261	3,262,663
ForeignInv_Telecom	276	2.8E+11	1.61E+08	2.32E+12	-2.4E+08	3.54E+13
Intl_Bandwidth	630	4,929,946	426,857.5	18,215,720	64.47	2.22E+08
Mobile_BB_Active	1184	37,545,646	6,490,055	1.42E+08	1846	1.87E+09
smartphone	129	83.02828	85	16.11933	8.90289	100
Access_Electricity	948	85.2481	99.9	24.77663	5.4	100
GDP_per_Capita	1079	19,494.09	6659.597	30,870.64	153.9302	256,580.5
GDP_perCapita_Growth	1078	1.405664	1.797616	5.925539	-30.4002	62.11107
Income_Inequality	382	35.3877	34.3	6.838203	24	55.1
Urban_Population	1106	60.07287	60.1835	23.05944	13.032	100
Account_Ownership	232	70.16036	73.10956	24.51523	5.831407	100
Broad_Money	746	64.85085	56.4429	40.34435	10.71918	284.261
Bank_Branches	798	15.46703	12.07585	12.65886	0.381991	92.00939
Private_Credit	923	54.33602	41.84067	42.21265	1.52681	220.3813
Digital_Payments	195	58.96986	57.66502	25.5859	4.81071	100
Digital_Accounts	80	47.68648	48.15184	21.32126	7.428223	96.20489
Merchant_Payments	156	25.55228	18.23376	21.45517	0	94.94146
Govt_Payments_Account	188	21.46174	15.52947	17.88328	0.562694	88.11368
Govt_Payments_Cash	188	3.816621	2.434434	4.110259	0	20.81265
Political_Stability	936	-0.0648	-0.01707	0.939415	-2.7769	1.669218
Regulatory_Quality	936	0.027841	-0.10576	0.952386	-2.30232	2.308591
Rule_of_Law	936	0.001496	-0.14142	0.959251	-2.13824	2.034399
Consumer_Protection	730	0.806849	1	0.39477	0	1
Cybersecurity_Framework	692	0.836705	1	0.369634	0	1
Data_Policies	143	0.601399	1	0.48961	0	1
Digital_Dev_Strategy	678	0.856932	1	0.350142	0	1
Digital_ID_Framework	606	0.541254	1	0.498295	0	1
Dispute_Resolution	728	0.976648	1	0.151018	0	1
EApps_Framework	643	0.437014	0	0.496017	0	1
Enforcement_Appeals	739	0.994587	1	0.073372	0	1
Foreign_Participation	714	0.284314	0	0.451087	0	1
ICT_Policies	693	0.897547	1	0.303243	0	1
Infra_Sharing	1014	0.60355	1	0.48916	0	1
Internet_Governance	719	0.200278	0	0.400208	0	1
Intl_Connectivity_Access	826	0.997579	1	0.049147	0	1
Intl_Facilities_Access	707	0.623762	1	0.484441	0	1
QoS_Regulation	740	0.954054	1	0.209368	0	1
Regulatory_Roadmap	150	0.64	1	0.48	0	1
External_Balance	961	-3.48624	-1.51316	15.88382	-95.7463	43.09438
Inflation	1017	7.897017	3.532361	26.06134	-12.297	557.2018
REER_Index	539	99.62353	99.03735	16.13076	53.79096	180.226

Beyond the macroeconomic environment, the Financial Depth and Inclusion pillar is the channel through which financial-stability considerations enter the index directly. The banking literature identifies deposit substitution, bank disintermediation, liquidity risk and the composition of bank funding as the central transmission mechanisms through which a CBDC affects financial stability (Bindseil 2020; Mancini-Griffoli et al. 2018). The indicators in this pillar operationalize the structural capacity to absorb these pressures: account ownership and broad money capture the scale and monetization of the deposit base, private credit and bank branches proxy the depth and reach of intermediation, and digital and merchant payment usage measure the extent to which transactional balances already circulate through digital rails rather than through bank deposits exposed to substitution. A financially deep and inclusive system can accommodate a partial migration of transactional balances into a CBDC with less disruption to bank funding, whereas a shallow system faces a sharper disintermediation margin. The pillar therefore embeds financial-stability resilience as a measured structural condition rather than treating it solely as a design-stage concern.

Institutions and Regulation. CBDC design and operation depend on legal mandates, regulatory quality, data governance and consumer protection (Cheng 2023; Nabilou 2020; Pocher and Veneris 2022; Soana and de Arruda 2024). The pillar combines the three World-wide Governance Indicators with binary flags for ICT strategy, digital-development and data-protection frameworks, cybersecurity regimes, digital ID, dispute resolution, quality-of-service regulation and connectivity mandates, reflecting evidence that implementation requires *ex ante* regulatory coherence rather than *ex post* patching. It captures formal institutional capacity, not public trust or political legitimacy, so high scores reflect implementation capacity rather than guaranteed adoption (Infante et al. 2022).

Two features warrant acknowledgement. First, several binary flags carry very high prevalence (Intl_Connectivity_Access 0.998, Enforcement_Appeals 0.995, Dispute_Resolution 0.977, QoS_Regulation 0.954) and therefore contribute limited cross-country discrimination under CRITIC, so effective weight concentrates on the three WGI variables and medium-prevalence flags; the parsimonious specification in Sect. 4.3 confirms the ranking is robust to dropping these flags. Second, geometric aggregation with binary flags compresses scores where a flag is zero; the safeguard epsilon of ten to the minus six prevents undefined values while preserving the intended non-compensatory behaviour. Implications for individual cases such as Spain are discussed in Sects. 4.5 and 5.1.

Macro Resilience and Policy. This pillar measures the macroeconomic environment in which a CBDC would operate. Open-economy models show that CBDC effects depend on initial conditions, including external imbalances, exchange-rate pressures and inflation credibility (Barrdear and Kumhof 2022; Karau 2023; Mishra and Prasad 2024; Moro and Nispi Landi 2024). External balance proxies exposure to currency substitution and capital-flow volatility, and inflation is treated as a cost indicator because monetary credibility shapes trust in public money. The real effective exchange rate is normalised relative to a long-run cross-country reference so that values near the reference score higher while pronounced deviations in either direction are penalised, since overvaluation signals competitiveness erosion and undervaluation signals external stress. These variables capture structural policy space within which design tools can operate credibly (Bindseil 2020; Auer et al. 2022) rather than predicting macro effects directly.

The index thus focuses on structural capacities and complements micro-level studies of adoption, trust and bank-level adaptation, which it does not measure directly. This scope is a deliberate design choice that keeps the index auditable and comparable across countries.

3.1.1 Preprocessing and normalization (per-year, robust)

Normalization is applied on a **per-year** basis so that index scores reflect contemporaneous relative standing. The pipeline implements the following deterministic chain:

- Winsorization at the 1st and 99th percentiles to limit extreme leverage:

$$x_{i,j,t}^{(w)} = \text{clip}\left(x_{i,j,t}, P_1^{(j,t)}, P_{99}^{(j,t)}\right).$$

- Robust standardization using the per-year median and the median absolute deviation (MAD).

Define $m_{j,t} = \text{median}_i\{x_{i,j,t}^{(w)}\}$ and $\text{MAD}_{j,t} = \text{median}_i\{|x_{i,j,t}^{(w)} - m_{j,t}|\}$. The robust z-score is:

$$z_{i,j,t} = \frac{x_{i,j,t}^{(w)} - m_{j,t}}{1.4826 \cdot \text{MAD}_{j,t}},$$

where the factor 1.4826 makes MAD comparable to the standard deviation under Gaussian assumptions. If $\text{MAD}_{j,t}$ is degenerate or non-finite the pipeline falls back to the mean absolute deviation, the fallback is recorded as a diagnostic.

- Direction alignment (cost variables). Indicators conceptually representing costs (for example inflation) are inverted prior to scaling so that, uniformly, larger standardized values indicate greater readiness.
- Clipping and rescaling. Robust z-scores are clipped to $[-3, 3]$ and linearly transformed to $[0, 1]$:

$$\tilde{v}_{i,j,t} = \frac{\text{clip}(z_{i,j,t}, -3, 3) + 3}{6}.$$

Binary policy flags remain in $\{0, 1\}$ and are not rescaled.

Preprocessing and normalization (per-year, robust) Normalization is applied on a per-year basis so that index scores reflect contemporaneous relative standing. Full normalization diagnostics and representative normalized distributions are provided in the online supplementary packages, organized by sample (Package 1: 183 sample; Package 2: 70 sample;

Package 3: 50 sample; Package 4: 40 sample; Package 5: BIS sample). These files report, for each variable and year, the median, MAD, number of observations, and any fallback events used in normalization. For transparency and reproducibility, the supplementary files also report per-year counts (n), percent missing, and the number of winsorized observations per variable.

The per-year structure handles variable-specific coverage contemporaneously, standardising each variable against the same-year cross-country distribution, and downstream CRITIC weighting further reduces the influence of low-dispersion indicators. The residual issue, that thinly covered countries are benchmarked against a partially different indicator set within a pillar, is acknowledged in Sect. 5.1.

3.1.2 Index construction, robustness and uncertainty quantification

The baseline index uses a two-stage CRITIC information weighting with geometric aggregation at both variable $\rightarrow$ pillar and pillar $\rightarrow$ index stages. The approach emphasizes indicators with high dispersion and low redundancy and avoids compensatory aggregation across pillars.

Within-pillar (CRITIC) weighting For a given pillar p and year t let $X_{t,p}$ denote the matrix of normalized (0–1) variables. Compute column standard deviations $s_{j,t}$ and the absolute pairwise correlation matrix $R_{jk,t} = |\text{corr}(X_{t,p}^{(j)}, X_{t,p}^{(k)})|$ (pairwise complete). The CRITIC information score is:

$$C_{j,t} = s_{j,t} \sum_{k \in p} (1 - R_{jk,t}),$$

and the within-pillar weights are:

$$w_{j,t} = \frac{C_{j,t}}{\sum_{j \in p} C_{j,t}}.$$

Equal weights are used as a documented fallback when the information vector is degenerate.

Pillar aggregation The pillar score for country i is the weighted geometric mean of the pillar's normalized components:

$$\text{Pillar}_{i,p,t} = \exp\left(\sum_{j \in p} w_{j,t} \ln(\max(\varepsilon, \tilde{v}_{i,j,t}))\right),$$

with ε a small positive safeguard (e.g. 10^{-6}) to avoid logarithms of zero.

Across-pillar weighting and overall index Pillar weights $W_{p,t}$ are computed using the same CRITIC logic applied to the cross-country dispersion and correlations of pillar scores in year t . The overall CBDC Readiness Index is

$$\text{Index}_{i,t} = \exp\left(\sum_p W_{p,t} \ln(\text{Pillar}_{i,p,t})\right).$$

We compute the index only when a country-year has at least three finite pillar scores; otherwise the country-year is flagged as missing.

Index construction and weighting Representative CRITIC weight allocations at the variable and pillar levels are reported in the supplementary packages for each sample. These outputs allow inspection of which variables or pillars dominate the index in specific years, while keeping the main manuscript focused on the conceptual pipeline and baseline results.

Robustness programme To ensure results are not artefacts of a single algorithmic choice, we estimate and compare multiple alternatives, with full outputs in the sample-specific supplementary packages (Packages 1–5): alternative within-pillar weights (equal, entropy, PCA), arithmetic in place of geometric aggregation, an orthogonal TOPSIS closeness-coefficient check, leave-one-pillar-out (LOPO) and leave-one-variable-out (LOVO) exercises with per-country rank deltas, and Spearman and Kendall concordance diagnostics across specifications.

Uncertainty quantification Two complementary procedures quantify sampling and measurement uncertainty, with complete outputs reported in the supplementary packages (Packages 1–5). First, a nonparametric bootstrap resamples the cross-section within each year ($B=500$, fixed seed) and recomputes weights and indices; we report rank mean, rank standard deviation, 5th and 95th percentile ranks, and empirical top-k probabilities (top-3, top-5, top-10). Second, a measurement-perturbation exercise performs $M=200$ draws where normalized inputs receive small, truncated Gaussian noise ($\sigma=0.02$), recomputes indices, and reports rank dispersion metrics and stability probabilities.

4 Index results

4.1 Results on base sample and base model

The index exhibits a right-skewed cross-country distribution: most economies cluster in lower-to-middle readiness bands while a small group attains substantially higher scores (Fig. 1). Coverage varies across economies (Fig. 2, 3), and the baseline CRITIC-weighted geometric index places small, highly interconnected financial jurisdictions and advanced economies at the top, with low-income economies that have limited digital infrastructure and weaker institutions in the bottom decile (Figs. 4 and 5). The leading positions of Bermuda and Monaco reflect the joint effect of strong per-capita digital and financial indicators, a scale effect examined directly in Sect. 4.3.

At the pillar level, Spearman correlations show systematic co-movement among Digital, Economic and Financial pillars, whereas Institutions and Regulation correlates only weakly with these three (Fig. 3). This supports the multi-pillar design: the index aggregates partially complementary capacities rather than collapsing to a single latent factor, and the relative independence of the Institutions pillar indicates that governance quality captures a distinct dimension that does not simply proxy development or digital penetration.

We assess rank reliability via a nonparametric cross-sectional bootstrap ($B=500$ draws, fixed seed) that resamples countries within each year, re-estimates CRITIC weights and recomputes the index (Table 3, Fig. 6). The highest-ranked jurisdictions are essentially invariant, with integer mean ranks, rank standard deviations near zero and top-5 probabilities of one, while economies near decile boundaries show appreciable dispersion: Viet Nam

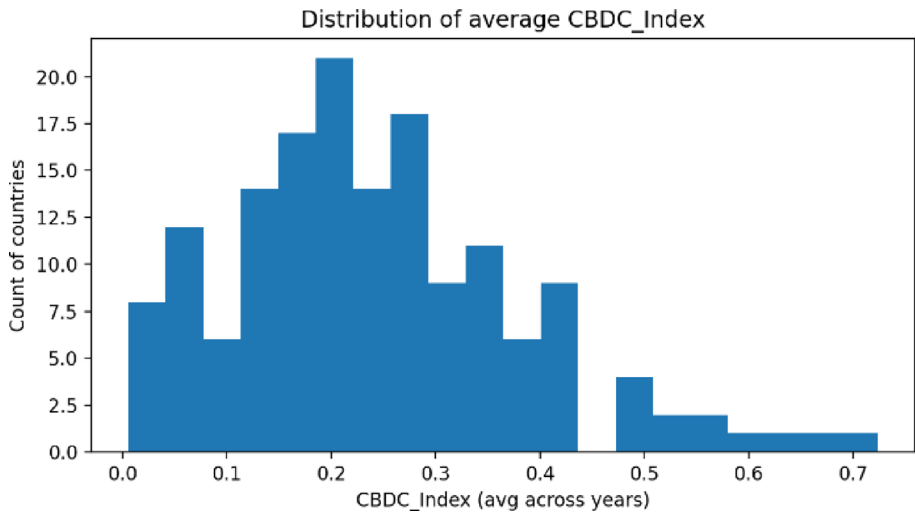


Fig. 1 Distribution of Average CBDC Index

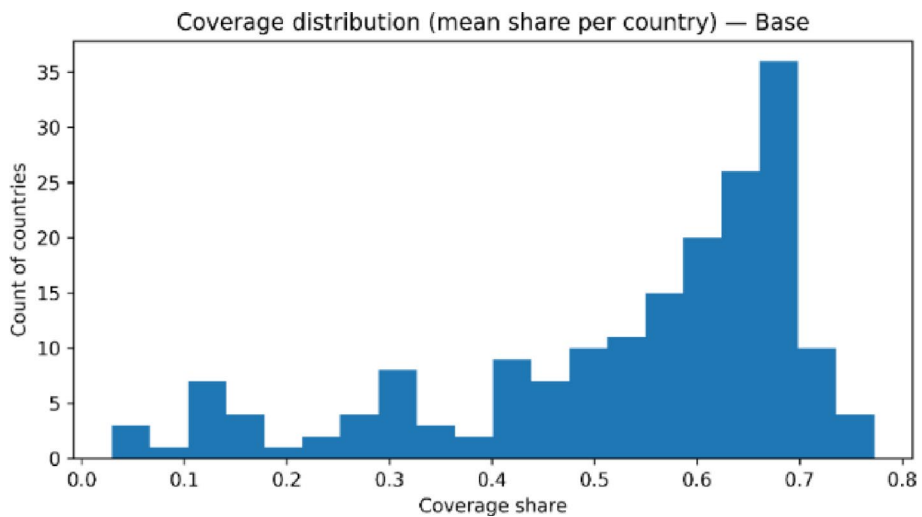


Fig. 2 Coverage Distribution

has a mean rank of about 10.12 with a standard deviation near 0.33 and a top-10 probability of about 0.876. Such borderline cases are scrutinised in the robustness exercises.

In interpretive terms, Fig. 6 shows that the vertical position of each economy, its mean bootstrap rank, is accompanied by a horizontal spread that is negligible at the top and bottom of the distribution and widest through the middle band. The reading is therefore that leader and laggard classifications can be taken as firm, whereas a difference of a few ranks among middle-band economies should be treated as within sampling noise rather than as a substantive ordering.

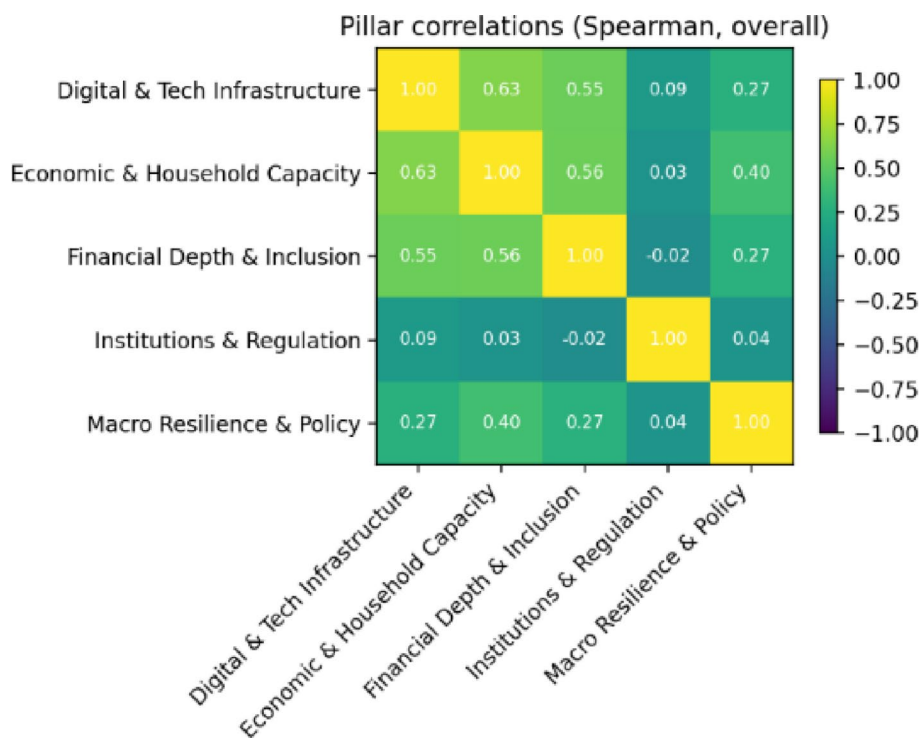


Fig. 3 Pillar Correlation

Taken together, the baseline evidence supports three conclusions: the index distinguishes distinct institutional, digital and financial configurations rather than a single development axis; leader and laggard groups are stable under resampling; and middle-band positions are appropriately sensitive, so the index should be read diagnostically rather than as a precise ordinal league table.

4.2 Cross-sample comparison results

We compare the baseline CBDC Readiness Index, computed on the full country panel (FULL_183), with alternative samples constructed by varying indicator coverage thresholds (SAMPLE_40, SAMPLE_50, SAMPLE_70) and a BIS-only subset (SAMPLE_BIS). The purpose of this exercise is to evaluate whether headline index levels and relative country orderings are sensitive to plausible changes in the sample frame and to identify which dimensions of readiness drive any differences.

4.2.1 Index-level concordance and coverage effects

Index-level distributions are qualitatively stable across samples. The full and coverage-restricted samples share the same broad shape (Fig. 7), and restricted samples lie near the identity line when plotted against FULL_183, indicating strong concordance in country scores (Fig. 8). Mean absolute Spearman correlations between pillar pairs are similar for

Top countries per metric

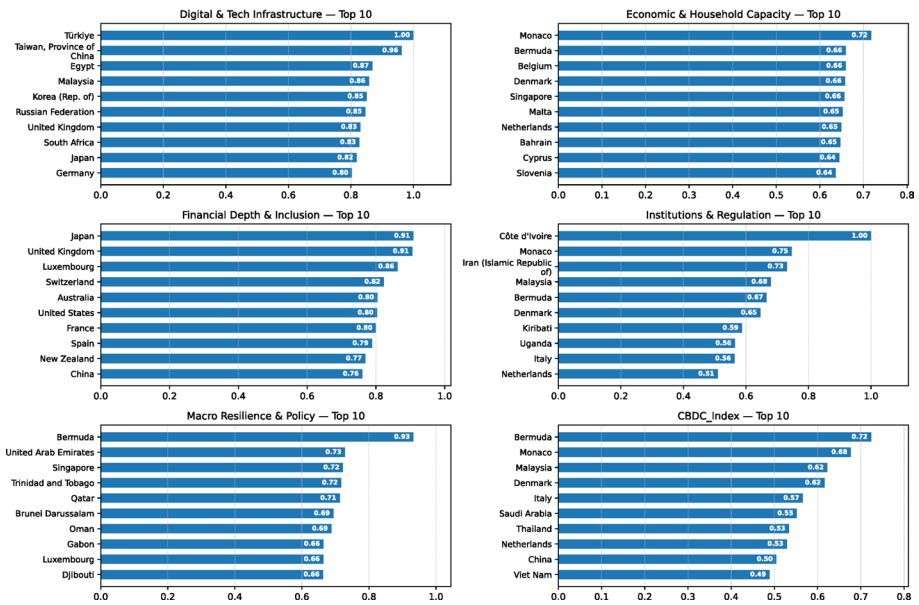


Fig. 4 Top 10 countries per pillar and overall (183 sample/ base index)

Bottom countries per metric

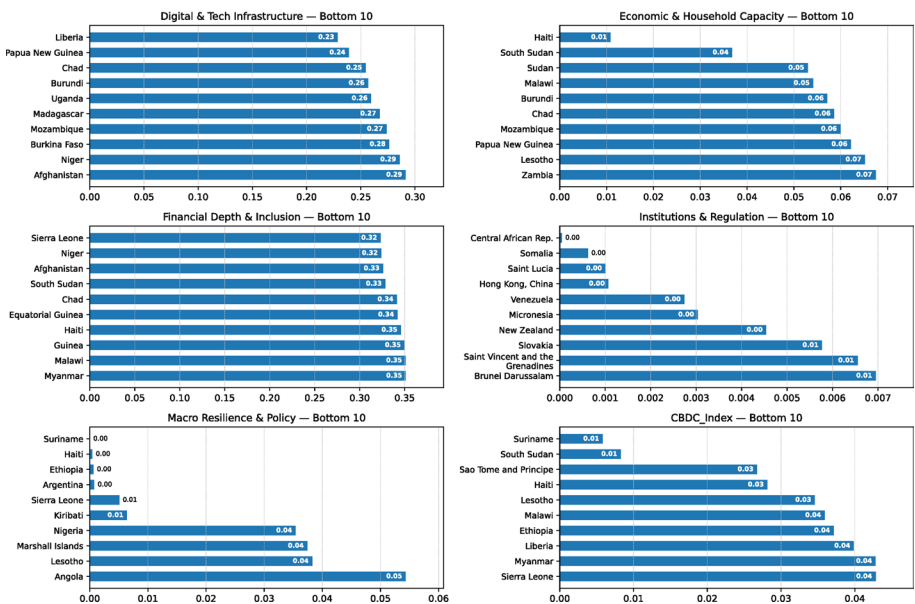
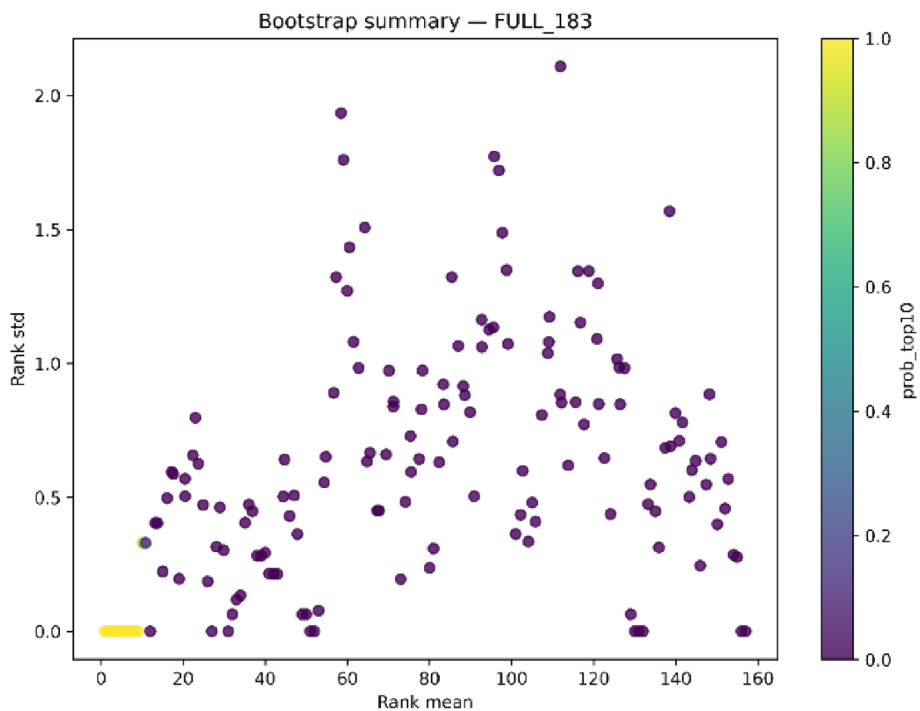


Fig. 5 Bottom 10 countries per pillar and overall (183 sample/ base index)

Table 3 Bootstrap Results (top 10 countries)

Country	Rank_mean	Rank_std	Rank_p5	Rank_p95	Prob_top3	Prob_top5	Prob_top10
Bermuda	1	0	1	1	1	1	1
Monaco	2	0	2	2	1	1	1
Malaysia	3	0	3	3	1	1	1
Denmark	4	0	4	4	0	1	1
Italy	5	0	5	5	0	1	1
Saudi Arabia	6	0	6	6	0	0	1
Thailand	7	0	7	7	0	0	1
Netherlands	8	0	8	8	0	0	1
China	9	0	9	9	0	0	1
Viet Nam	10.124	0.329582	10	11	0	0	0.876

**Fig. 6** Bootstrap summary plot

FULL_183, SAMPLE_40, SAMPLE_50 and SAMPLE_70 (about 0.33–0.345) and only moderately lower for SAMPLE_BIS (about 0.27), consistent with its narrower indicator set (Fig. 9). Coverage variation therefore affects dispersion and mean levels rather than the cross-country ordering.

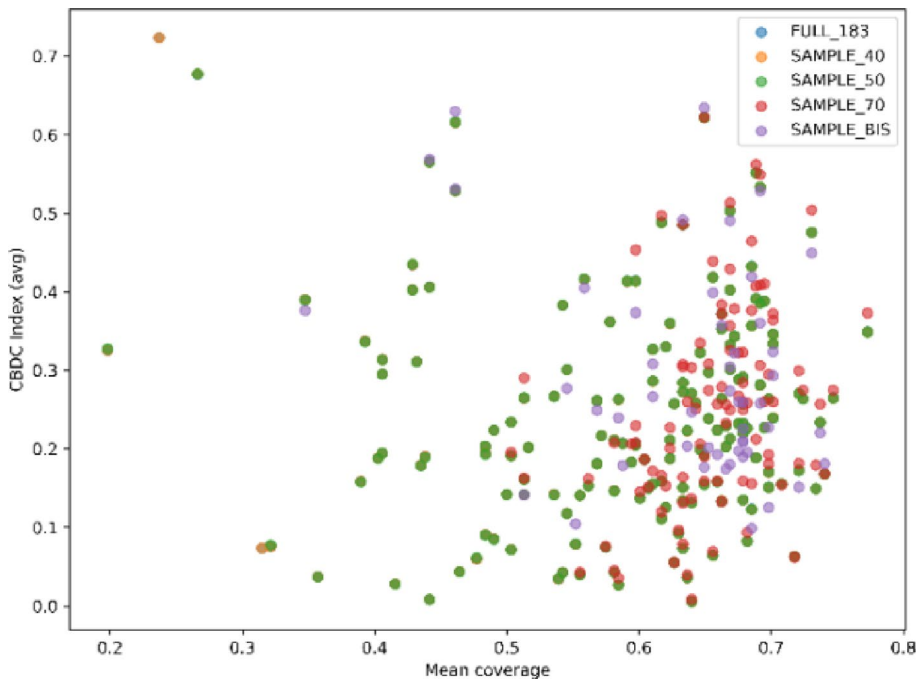


Fig. 7 CBDC Index (avg) vs mean coverage, cross-sample scatter

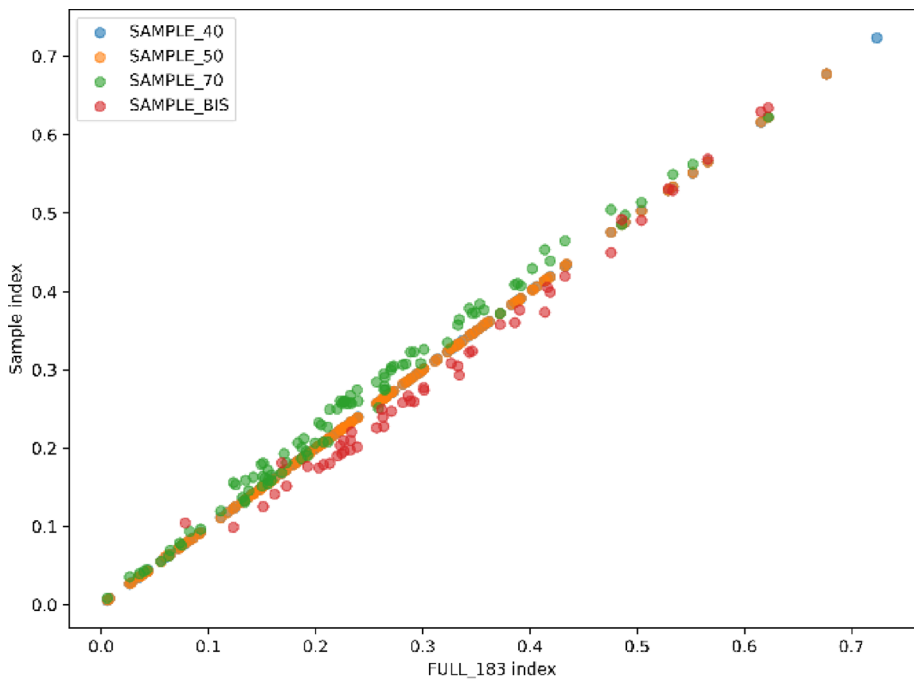


Fig. 8 Sample indices plotted against FULL_183 index (identity line for reference)

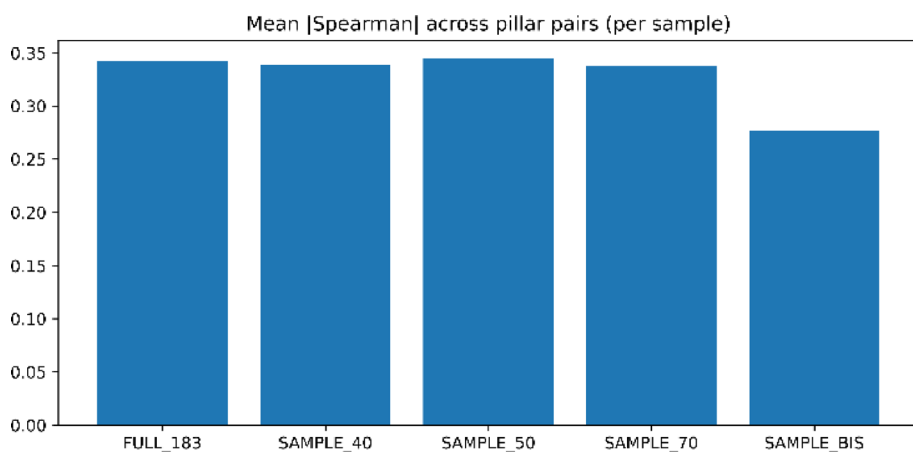


Fig. 9 Mean |Spearman| across pillar pairs by sample

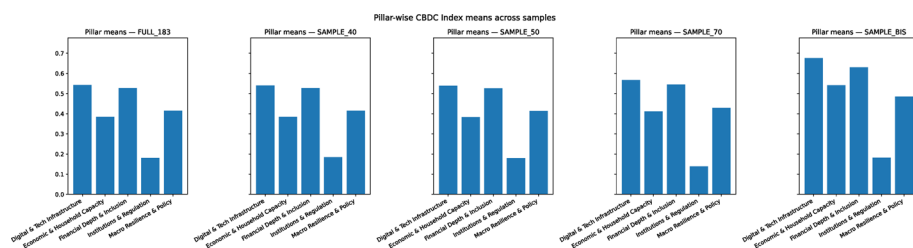


Fig. 10 Pillar-wise CBDC Index means across samples

4.2.2 Pillar-level shifts and interpretation

Sample differences are most pronounced at the pillar level (Fig. 10). The BIS subset and high-coverage SAMPLE_70 show higher means in the Digital and Financial pillars, reflecting composition effects that weight indicators well measured for advanced, connected economies. These shifts change the magnitude of pillar gaps but not the identity of the leader and laggard groups, so policy inference about which dimensions require reform remains stable.

4.2.3 Top and bottom decile stability and rank uncertainty

The composition of the top and bottom deciles is robust across samples (Figs. 11 and 12). Bootstrap rank intervals show leading countries with near-zero dispersion and top-5 or top-10 probabilities close to one, while borderline cases such as Viet Nam (mean rank about 10.12, standard deviation about 0.33, top-10 probability about 0.876) warrant guarded interpretation (Fig. 13, Table C). Two conclusions follow: the CRITIC-weighted geometric procedure yields a stable ranking under plausible coverage variation, and sample choice affects pillar-level magnitudes rather than the broad leader and laggard structure. We next turn to an expanded robustness programme covering alternative weighting, aggregation and leave-one-out experiments.

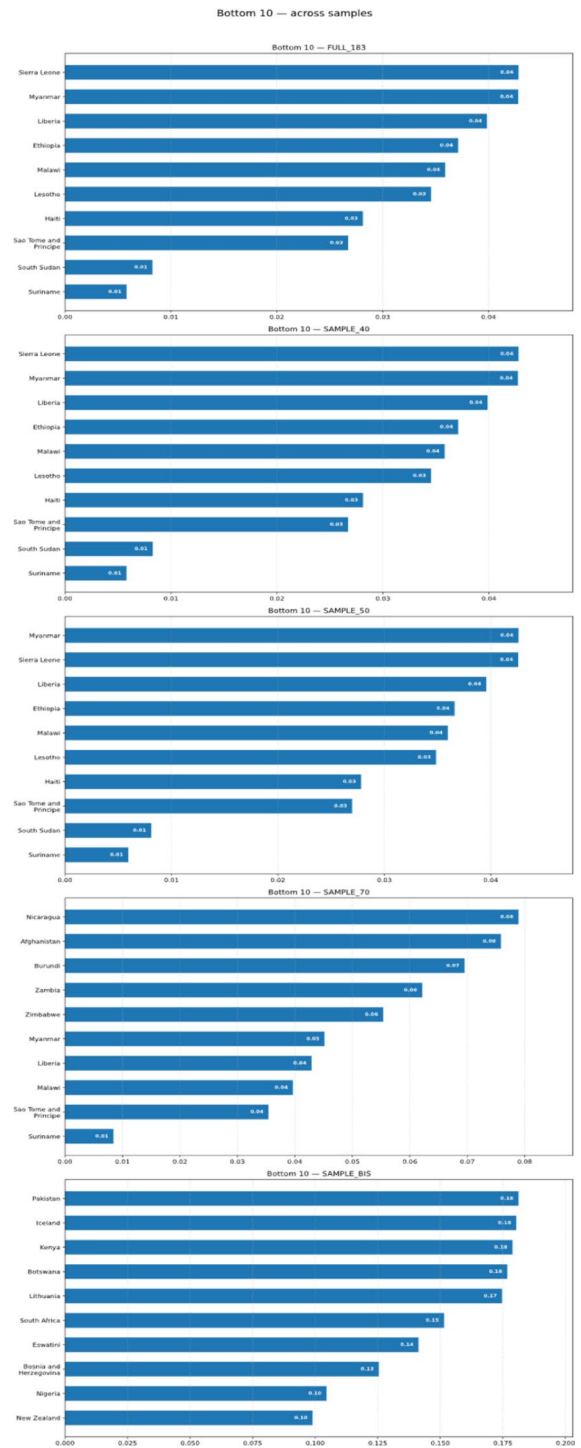
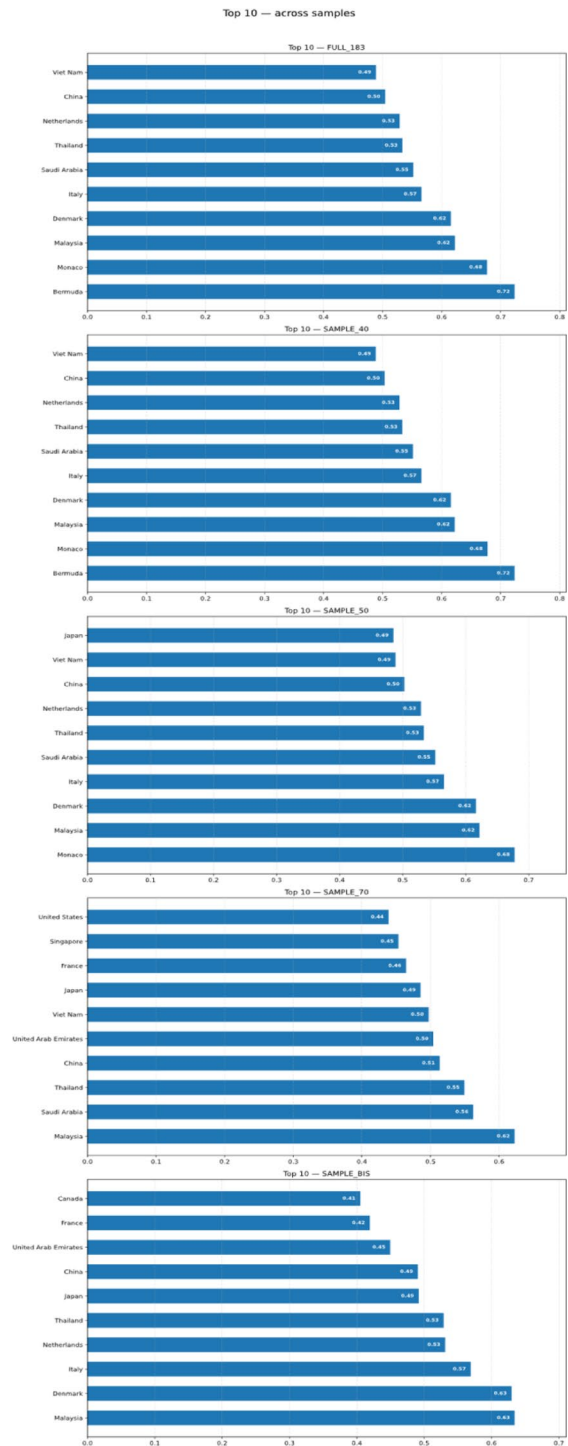
Fig. 11 Bottom-10 composition across Samples

Fig. 12 Top-10 composition across samples



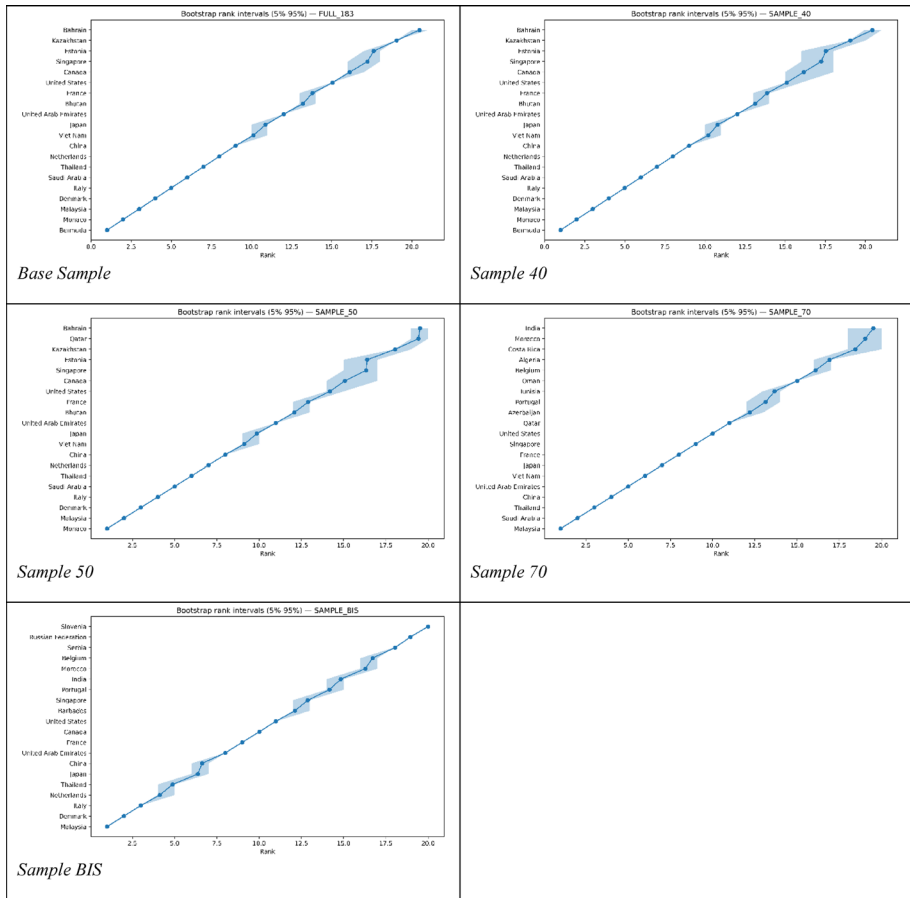


Fig. 13 Bootstrap rank intervals (5–95%) for selected ranks across the five samples

4.3 Robustness and sensitivity results

We evaluate stability through a systematic battery of robustness and sensitivity checks: bootstrap rank-displacement diagnostics, concordance across aggregation rules and sample frames, leave-one-variable-out and pillar-exclusion experiments, and model comparison of top-rank stability. Each is reported below, with full outputs in the supplementary packages.

4.3.1 Bootstrap rank-displacement diagnostics (global stability)

We quantify rank uncertainty using a nonparametric cross-sectional bootstrap and summarise absolute rank changes, $|\Delta \text{rank}|$, across samples and aggregation variants (Figs. 14, 15 and 16). Median absolute rank changes are small for top and bottom economies and larger in the middle of the distribution, confirming that headline positions are stable while intermediate ranks carry measurable uncertainty.

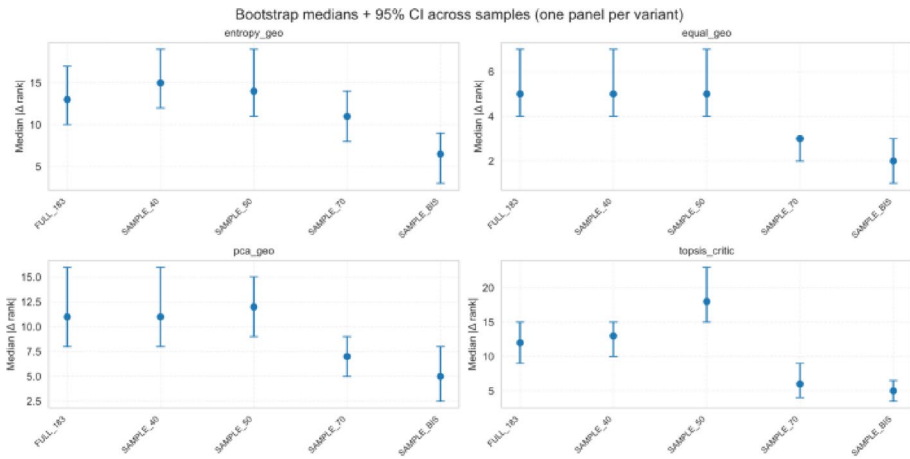


Fig. 14 Bootstrap medians and 95 percent

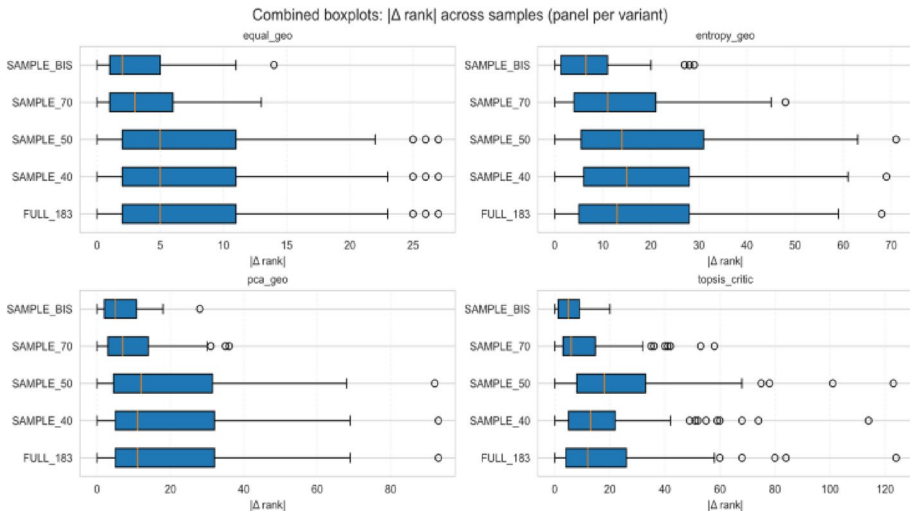


Fig. 15 Boxplot distributions of absolute rank changes by sample and aggregation variant

Per-variant histograms (Figs. 17, 18, 19 and 20) show that entropy and TOPSIS variants generate right-skewed rank-change distributions concentrated in the middle ranks, while equal-weight and PCA variants track the baseline closely.

4.3.2 Aggregation and sample sensitivity (concordance and pairwise agreement)

We compare rankings across aggregation rules and sample frames (FULL_183, SAMPLE_40, SAMPLE_50, SAMPLE_70, SAMPLE_BIS) using Kendall tau and Spearman rho (Figs. 21 and 22). Pairwise agreement is high across specifications, with the lowest values involving the narrower BIS subset.

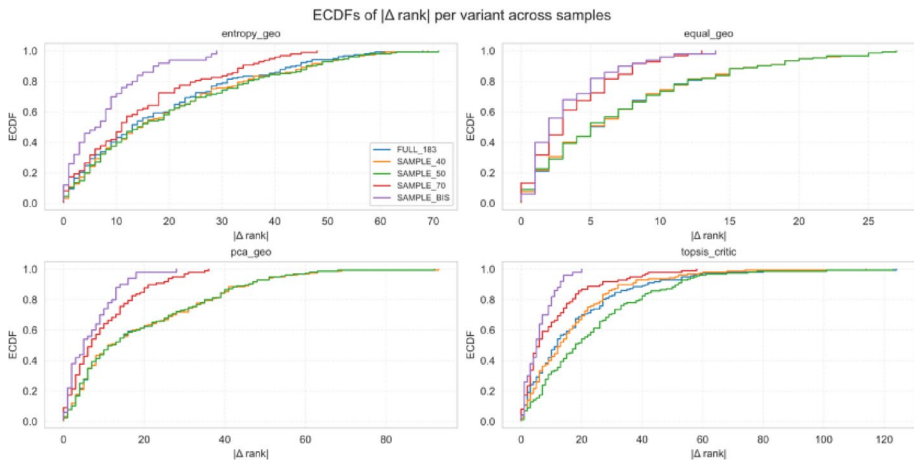


Fig. 16 Empirical cumulative distribution functions (ECDFs) of $|\Delta \text{rank}|$ across samples

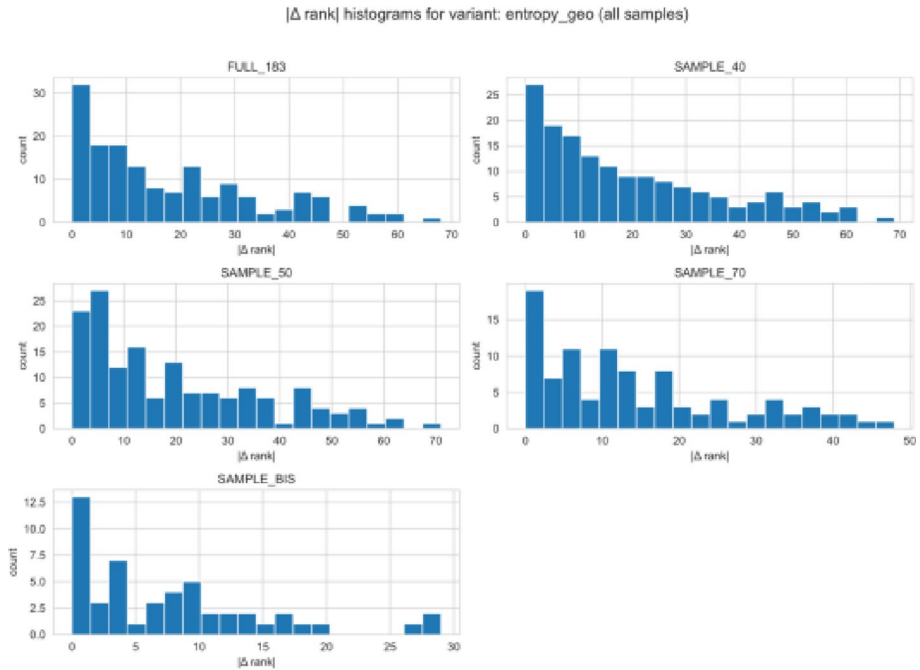


Fig. 17 $|\Delta \text{rank}|$ histograms for variant *entropy_geo* (all samples)

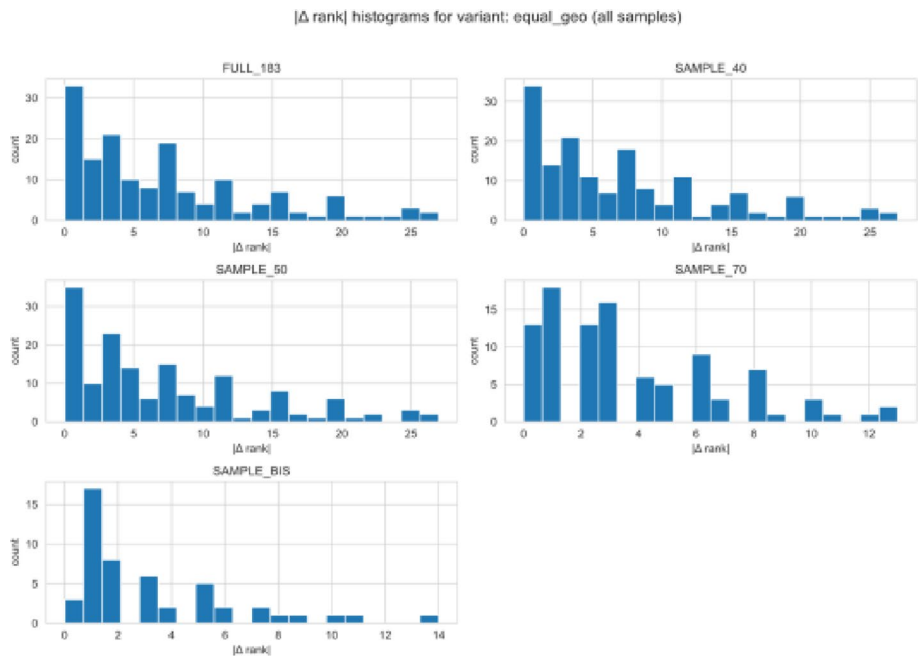


Fig. 18 $|\Delta \text{rank}|$ histograms for variant *equal_geo* (all samples)

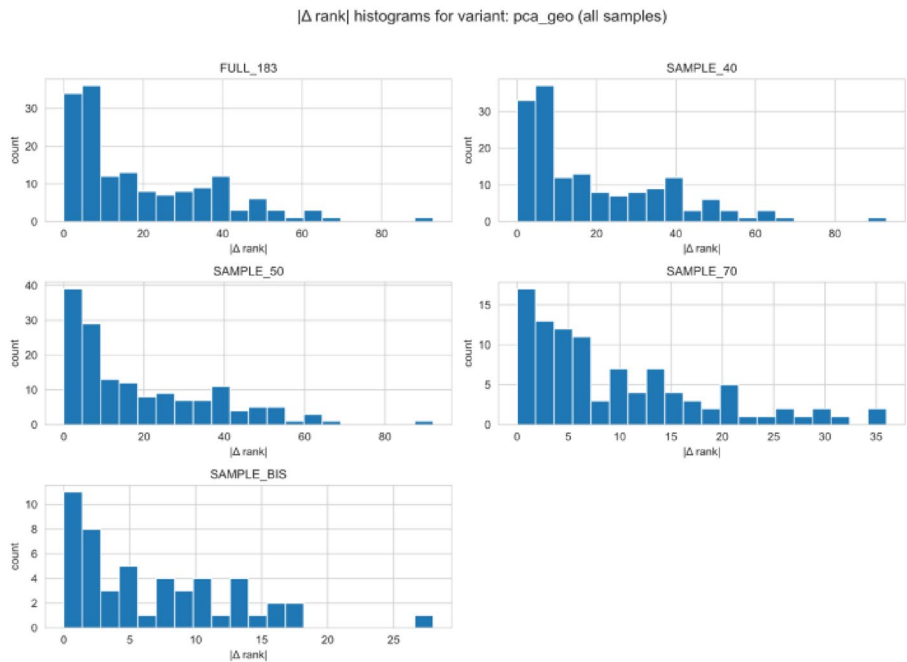


Fig. 19 $|\Delta \text{rank}|$ histograms for variant *pca_geo* (all samples)

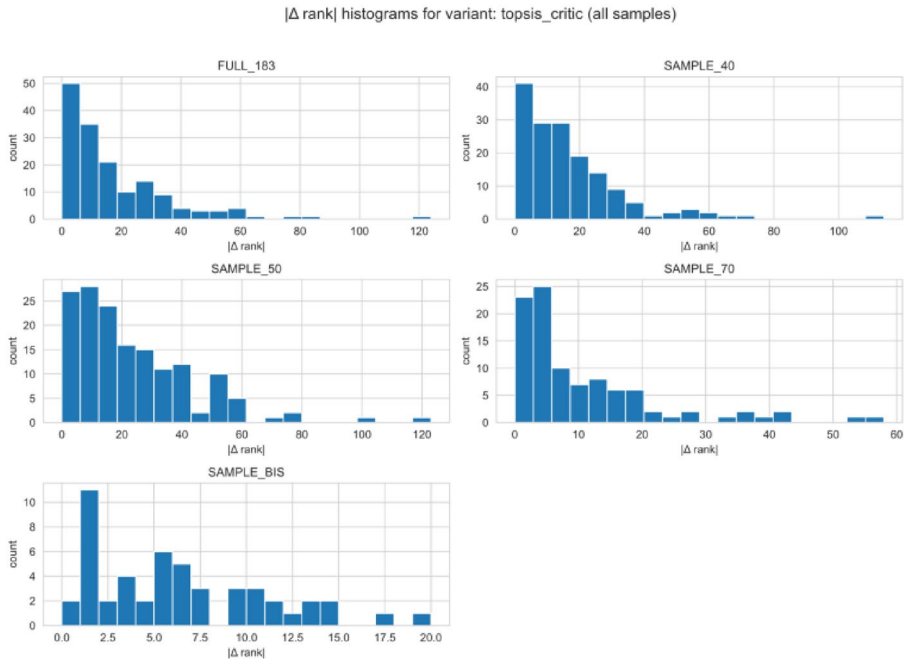


Fig. 20 | Δ rank| histograms for variant *topsis_critic* (all samples)

Violin plots with box overlays (Fig. 23) present the full distribution of |delta rank| across samples and variants, reinforcing that most economies move only a few ranks under reasonable specification changes.

4.3.3 Indicator- and pillar-level sensitivity (mechanisms of instability)

To identify which elements drive rank volatility we ran leave-one-variable-out (LOVO) and pillar-exclusion experiments (Figs. 24 and 25). LOVO isolates indicators whose omission causes the largest average rank shifts, and the pillar-exclusion test quantifies the effect of dropping each conceptual domain.

Read together, Figs. 24 and 25 localize where rank volatility originates. The leave-one-variable-out panel identifies a small number of high-leverage indicators within the digital and financial pillars whose removal produces the largest average rank shifts, while the pillar-exclusion panel shows that dropping the Institutions and Regulation pillar moves the middle of the distribution more than the extremes. The interpretation is that headline positions are driven by broad pillar structure rather than by any single indicator, which is the property a structural index should display.

The results are clear in direction. A small set of indicators in the Digital and Financial pillars carries the most leverage, and dropping the Institutions and Regulation pillar moves the middle of the distribution more than the extremes. Headline positions are therefore driven by broad pillar structure rather than by any single indicator, the property a structural index should display.

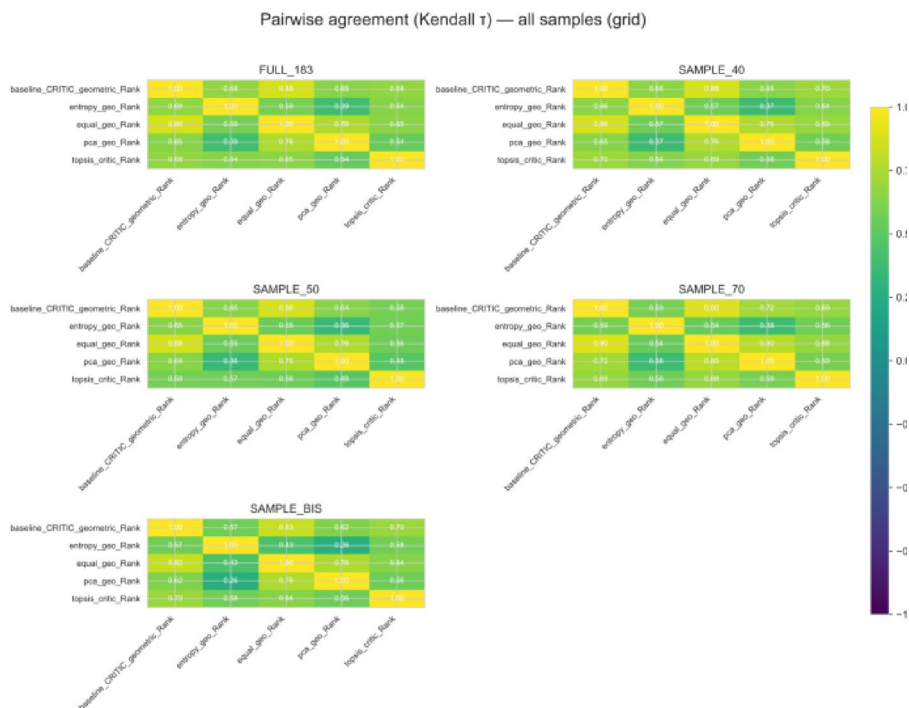
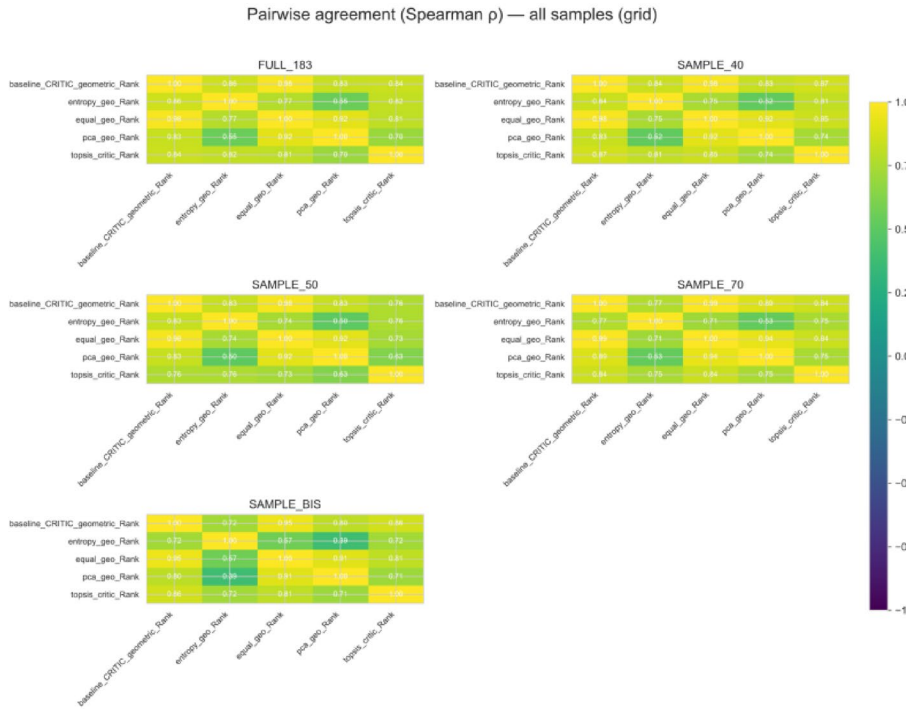
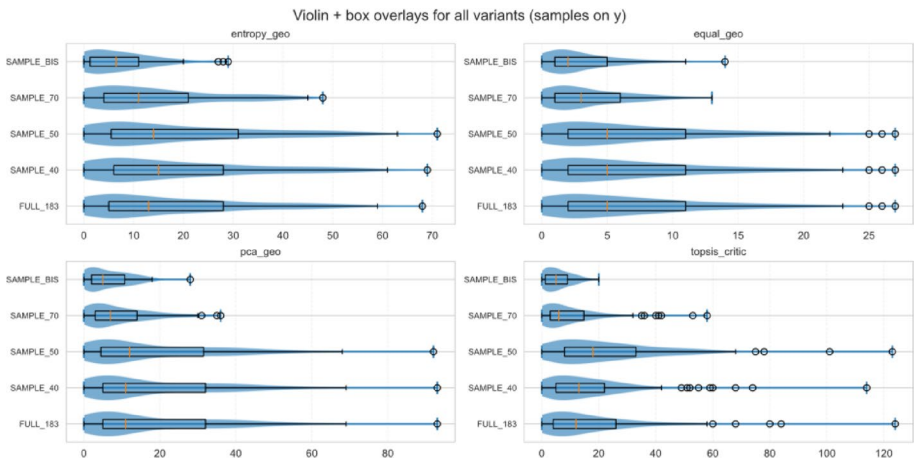


Fig. 21 Pairwise rank agreement (Kendall τ)

The qualitative picture is consistent: across every alternative weighting (equal, entropy, PCA), every aggregation rule and every coverage sample, the top and bottom groups remain essentially fixed while intermediate ranks shift within bounded ranges. The substantive policy messages do not depend on a single algorithmic choice.

Two further sensitivity exercises address the influence of jurisdiction scale and of the weighting and aggregation choices on the ranking; both are reported in full in the Robustness Results Appendix (Appendix R, Sections R.2 and R.4). Per-capita normalization can elevate small jurisdictions with offshore financial structures, because several digital and financial indicators scale inversely with population, and the two highest baseline ranks, Bermuda and Monaco, illustrate this mechanism. Recomputing the ranking after excluding the 26 economies with a population below one million leaves the leading positions occupied entirely by substantial national economies, including Malaysia, Denmark, Italy, the Netherlands, Saudi Arabia, Thailand, Viet Nam, the United Arab Emirates, China and Japan. The rank correlation between the baseline and the restricted ranking across all retained economies is unity, which confirms that micro-jurisdictions are removed cleanly without reordering the remaining cross-section. Reporting leaders within population tiers conveys the same conclusion: within every size band the leaders are economies with strong digital and financial fundamentals, so the substantive policy contest is preserved once scale is held comparable (Appendix R, Table R3, Figures R4 and R5). Population data are obtained from the World Bank (indicator SP.POP.TOTL, 2023).

Fig. 22 Pairwise rank agreement (Spearman ρ)Fig. 23 Violin plots with box overlays for $|\Delta \text{rank}|$

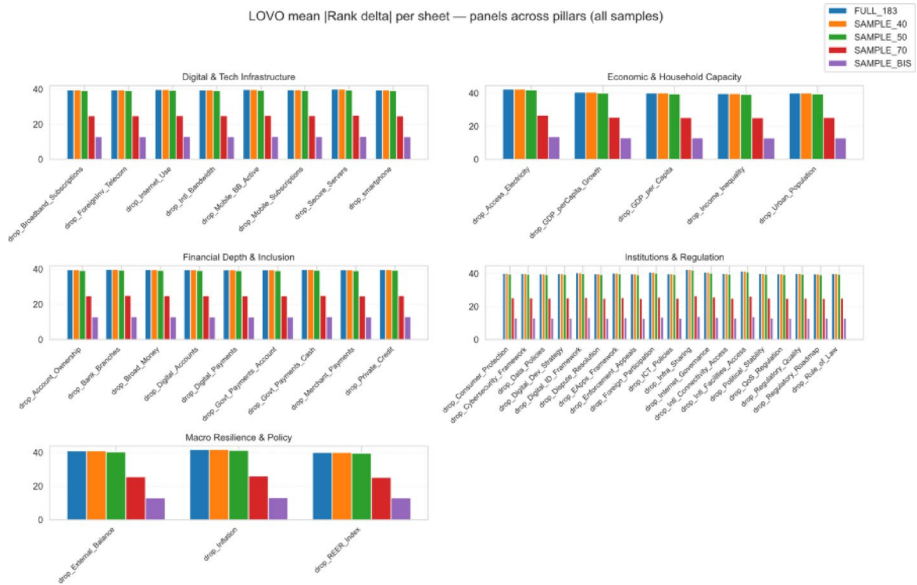


Fig. 24 Leave-one-variable-out (LOVO) mean |rank delta| per sheet (panels across pillars and samples)

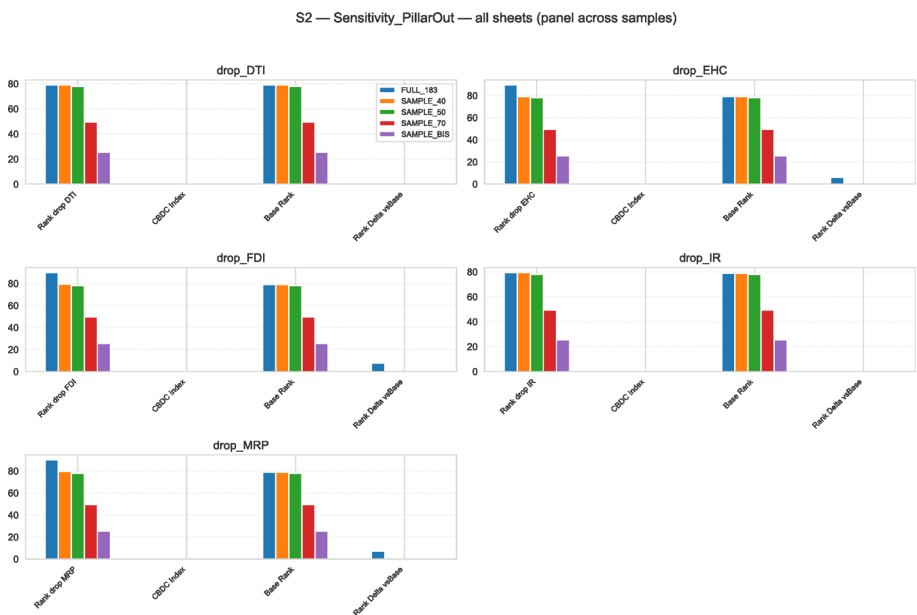


Fig. 25 Sensitivity exercise: Pillar-out results across sheets (panel across samples)

The weighting and aggregation choices are also probed beyond the data-driven CRITIC scheme. A theory-based weighting that assigns higher *a priori* weight to the Financial Depth and Institutions pillars, reflecting their centrality in disintermediation and credibility channels, reproduces the CRITIC ranking at a Spearman coefficient of 0.990, so the headline ordering is not an artefact of the information-weighting rule. A compensatory arithmetic aggregation correlates with the geometric baseline at a Spearman coefficient of 0.670, and the economies whose ranks rise most under this rule are precisely those whose only marked weakness is the institutionally depressed pillar, which confirms that the non-compensatory geometric rule is the operative source of separation between digital-financial strength and institutional codification. A rank-averaged hybrid of the two rules correlates with the baseline at 0.913. Finally, a parsimonious specification of the Institutions and Regulation pillar that removes the four binary flags with prevalence above 0.95, namely dispute resolution, enforcement and appeals, international connectivity access, and quality-of-service regulation, reproduces the baseline ranking at a Spearman coefficient of 0.997, demonstrating that the high-prevalence flags do not drive the results and that the pillar can be read as a governance-anchored score augmented by a small set of discriminating flags (Appendix R, Table R5, Figures R7 and R8).

4.4 Model comparison and top-20 stability

To complement rank-change diagnostics, model-comparison panels plot model-average scores for top-ranked economies across samples (Fig. 26). The top-20 set is highly stable: leaders retain their positions across weighting and aggregation variants, with only minor reordering within the group, confirming that the headline league of high-readiness economies is not an artefact of the baseline specification.

4.5 From index to toolkit: concrete policy application

To demonstrate the practical utility of the CBDC Readiness Index as a diagnostic toolkit rather than a mere ranking exercise, we compare pairs of countries with similar GDP per capita but divergent readiness profiles. The pairs are selected on three transparent criteria: (i) similar income status measured by average GDP per capita over the sample window; (ii) shared geographic region or shared monetary regime; and (iii) salience in the policy debate, that is, both jurisdictions appearing in the Atlantic Council CBDC Tracker as of 2025. Index scores in this subsection are reported to two decimal places to be consistent with the rank uncertainty bounds documented in Sect. 4.1, and ordinal interpretation should be guided by the bootstrap rank intervals already reported in Table 3 rather than by index point estimates alone. The comparisons draw on cross-year averages from the baseline CRITIC-geometric model and raw variable data, highlighting how pillar-specific strengths and weaknesses inform tailored policy sequencing and CBDC design choices.

The toolkit is illustrated with four income-matched country pairs whose readiness profiles diverge. Thailand scores well above Indonesia overall (0.53 vs 0.30), driven by stronger Financial Depth and Inclusion (0.75 vs 0.37, reflecting account ownership near 96 percent against 49 percent) and Digital infrastructure, while Indonesia lags in Institutions and Regulation; this matches the CBDC Tracker, where Thailand completed a retail pilot in 2024 (Bank of Thailand 2021) and Indonesia remains in research on Project Garuda. Italy

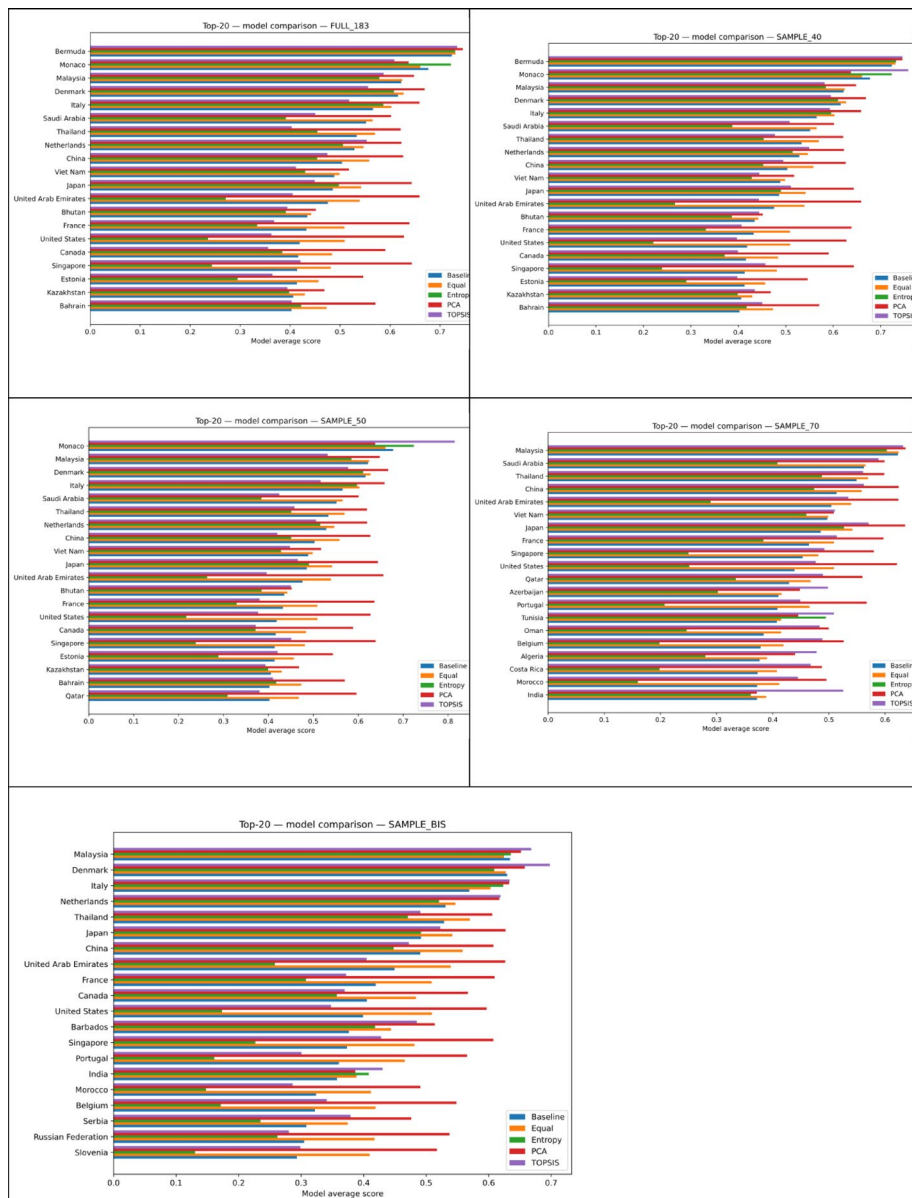


Fig. 26 Model comparison and top-20 stability across samples

outranks Spain among euro-area economies, with Spain constrained mainly by digital-governance flags (European Central Bank 2025). Kenya outranks Kiribati among small economies through mobile-money depth and digital reach, with Kiribati limited by infrastructure and missing data. Nigeria, despite launching the eNaira, scores low on Macro Resilience and institutions, while Ghana edges higher on digital and institutional flags.

The diagnostic translates into differentiated sequencing. Thailand's balanced profile supports moving toward a retail CBDC with institutional upgrades in data protection and cybersecurity, using tiered remuneration to leverage high digital-payment usage while limiting disintermediation (Maryaningsih et al. 2022). Indonesia should prioritise financial inclusion and institutional flags before piloting, focusing on wholesale to reduce substitution risk (Agur et al. 2019). Italy can pursue advanced euro-area designs after macro monitoring (Brunnermeier and Landau 2022), whereas Spain should first align digital-governance flags with EU standards. Kenya can build an inclusion-focused CBDC on existing mobile rails with offline support for rural areas (Das et al. 2023), while Kiribati must address basic digital and household-capacity bottlenecks before any CBDC, and Nigeria's low eNaira adoption underscores institutional reforms in rule of law and data policy before scaling (Osakwe et al. 2025; Illes et al. 2025).

4.6 External validation and construct validity

A structural readiness index must demonstrate that it captures conditions specific to central bank digital currency rather than reproducing a general development gradient. We establish this through a construct-validity exercise that links the index to independent external criteria drawn from the project, search-interest and central-bank-speech dataset of Auer, Cornelli and Frost (2020). The merged sample comprises 134 jurisdictions for which both a readiness score and an external project score are available. Full tables and figures are reported in the Robustness Results Appendix (Appendix R, Section R.1, Tables R1 and R2, Figures R1 to R3).

Convergent validity is supported by a positive and statistically significant association between the readiness index and independent measures of CBDC activity. The index correlates with the overall project score at a Spearman coefficient of 0.327, with retail and wholesale project scores at 0.218 and 0.254 respectively, and with normalized public search interest at 0.296. Jurisdictions with an active project record a mean readiness score of 0.283 against 0.176 for those with no recorded project. The association with the central-bank speech-stance index is small and statistically insignificant, which is consistent with the position that structural readiness is distinct from rhetorical or political posture. At the pillar level, the Digital and Technology Infrastructure pillar is most strongly associated with project advancement (Spearman 0.517), followed by Financial Depth and Inclusion and Economic and Household Capacity, while the Institutions and Regulation pillar is the weakest correlate. These magnitudes are moderate rather than mechanical, indicating convergence without redundancy.

Discriminant validity is the more demanding requirement. If the index merely restated economic development, it would carry no information about CBDC activity once income is held constant. Controlling for the logarithm of GDP per capita, the partial rank correlation between the index and the project score remains positive and significant at 0.206. In a nested ordinary-least-squares specification, GDP per capita alone and the readiness index alone each explain 12.5 percent of the variance in project stage, while the two jointly explain 17.9 percent, with the index retaining a significant coefficient. The index therefore contributes explanatory power beyond income, supporting the interpretation that it measures CBDC-specific structural capacity rather than development alone. We note that the external criteria reflect a snapshot of project activity, so the exercise establishes concurrent construct validity

rather than dynamic prediction; a forward-looking predictive design that tracks subsequent project transitions is identified in the future research agenda.

4.7 Readiness archetypes

Ordinal ranks compress a multidimensional profile into a single position and can obscure the configuration of strengths and weaknesses that matters for policy sequencing. To complement the ranking, we group economies by the shape of their pillar profile using k-means clustering on standardized country-average pillar scores. A four-cluster solution is retained because it aligns with interpretable readiness types and provides actionable granularity; the silhouette coefficient is 0.272 at four clusters, while a two-cluster partition maximizes silhouette at 0.325. Cluster profiles and membership are reported in Appendix R, Section R.3, Table R4 and Figure R6.

The first archetype consists of balanced high-readiness economies that score strongly across all five pillars, including Malaysia, Denmark, Italy, the Netherlands, China and Japan. The second comprises digitally and financially advanced economies whose Institutions and Regulation pillar score is depressed, including the United States, Singapore, Canada, France and the United Arab Emirates; as the robustness analysis shows, this depressed score reflects the interaction between high-prevalence binary regulatory flags and non-compensatory aggregation rather than a substantive governance deficit. The third archetype contains macro-stable but digitally and financially shallow economies such as Indonesia, Kenya and Bangladesh. The fourth contains structurally constrained economies with low scores across most pillars. Because each archetype implies a distinct binding constraint, the typology translates more directly into sequencing priorities than ordinal position: the second archetype points to regulatory codification, the third to digital and financial deepening, and the fourth to foundational infrastructure and macro stabilization.

5 Discussion and implications

The index portrays structural capacity for CBDC as highly uneven and multidimensional. Most economies sit in a broad middle band, while a small set combine strong digital infrastructure, deep inclusive financial systems, robust institutions and solid macro fundamentals. Pillar correlations show that Digital, Economic and Financial capacities move together, whereas Institutions is only weakly correlated with them and Macro adds a partly independent layer, so readiness is a bundle of partly distinct capabilities rather than a single development gradient.

These patterns help rationalise heterogeneous CBDC experiences. Models show that CBDC can stabilise or destabilise intermediation depending on design and underlying resilience (Andolfatto 2021; Chiu et al. 2023; Nyffenegger 2024). Our results suggest favourable and adverse outcomes emerge at opposite ends of the readiness distribution: high-readiness jurisdictions have more space for ambitious designs, while low-readiness countries face tighter constraints and larger downside risks if a CBDC is introduced before structural bottlenecks are addressed.

The pillar decomposition also clarifies the link between structural readiness and micro-level adoption: capacity is necessary but not sufficient for uptake, which additionally depends on trust and privacy perceptions outside the index.

Relative to other CBDC indices, the readiness index is deliberately structural and slow-moving. News-based and sentiment indices are best read as high-frequency signals and the readiness index as a slow-moving conditioning variable, so a given sentiment shock should have larger real effects in higher-readiness jurisdictions. Stage and maturity indices (Nguyen et al. 2025; PwC 2023) track the project pipeline but are driven by discrete, self-reported decisions that reflect political priorities as much as capacity. Structural readiness and project-stage advancement are therefore related but distinct, as the validation in Sect. 4.6 confirms.

The index makes explicit the preconditions implicit in IMF, BIS and central-bank frameworks, providing an auditable structural benchmark that researchers can use to condition adoption studies and event analyses, reducing omitted-variable bias when interpreting why identical announcements produce heterogeneous effects.

Operationally, the framework is applied in four steps: compute pillar and overall scores using the public data pipeline and CRITIC-geometric method of Sect. 3; identify binding constraints from relative pillar scores and key raw indicators; map gaps to sequencing priorities, addressing foundational weaknesses before design calibration; and stress-test proposed architectures against the readiness profile to calibrate scope and complementary reforms.

Finally, the robustness programme indicates that the main policy messages are not artefacts of a particular weighting or aggregation choice. Leader and laggard classifications, the multidimensional structure of readiness, and the archetypes are stable across alternative specifications, the exclusion of micro-jurisdictions, theory-based weighting and a parsimonious institutional pillar, which supports using the index as a diagnostic rather than an ordinal device.

5.1 Limitations

The index is constructed as a structural diagnostic of foundational conditions for CBDC implementation. We acknowledge several limitations that bound the scope of inference and that we treat transparently rather than obscure.

Several issues that bounded earlier versions are addressed in this revision and reported as results: construct validity against external criteria (Sect. 4.6), micro-jurisdiction influence through population-threshold and size-tiered rankings, theory-based weighting, compensatory and parsimonious specifications, and the archetype analysis. The genuinely open limitations are set out below.

Association with observed project stages. The index captures slow-moving fundamentals rather than discrete decisions to research, pilot or launch, so it is not designed to predict announcement timing or a country's position in the Atlantic Council CBDC Tracker.

The concurrent validation reported in Sect. 4.6 partially addresses this point by establishing that readiness is positively and significantly associated with observed project advancement while remaining distinct from economic development. What remains outside the present scope is the dynamic question of whether readiness predicts subsequent transitions in project stage, which requires a panel of project outcomes observed after the readiness measurement and is reserved for future work.

Micro-state interpretation. Per-capita normalisation can elevate very small jurisdictions with offshore financial structures; Sect. 4.3 addresses this through population-threshold exclusion and size-tiered rankings, but readers should still interpret micro-jurisdiction scores with reference to scale.

Coverage asymmetry. Indicator availability is uneven across countries within a pillar; per-year normalisation handles this contemporaneously, and the coverage-restricted samples in Sect. 4.2 show that realistic coverage variation does not reorder the cross-section.

Binary flags under geometric aggregation. Combining continuous WGI indicators with binary flags under a non-compensatory rule can compress scores where a flag is zero; this is a deliberate feature, and the parsimonious specification in Sect. 4.3 confirms the ranking is robust to removing near-constant flags.

WGI dominance. Because several flags have prevalence above 0.95, effective institutional weight concentrates on the three WGI variables; the flags are retained for conceptual completeness and signalling baseline regulatory presence.

Theory-led selection with data-led weighting. Variables are chosen on conceptual grounds while CRITIC sets weights from dispersion and correlation; the theory-weighted variant in Sect. 4.3 shows the ranking is not an artefact of this rule.

Toolkit country pairs. The four pairs in Sect. 4.5 are purposive illustrations, not a formal matching estimator; a matched-pair design is identified as future work.

CRITIC weight stability. Coverage composition can move annual CRITIC weights; a year-by-year characterisation with bootstrap intervals on each weight is left to future work.

Temporal dynamics. The main text emphasises cross-year averages; a systematic convergence analysis of the per-year indices is reserved for future work.

Behavioural and political dimensions. The index measures structural capacity, not behavioural trust, political legitimacy or specific design quality, which require micro-level data.

5.2 Future research

The revision completes several items that were previously identified as future work. External validation against independent CBDC project, search-interest and central-bank-communication data is reported in Sect. 4.6; population-threshold and size-adjusted robustness is reported in the Robustness Results Appendix; theory-based weighting, compensatory aggregation and a parsimonious institutional specification are reported alongside the baseline; and a clustering of economies into readiness archetypes is provided. The agenda that remains open is accordingly narrower and centers on dynamic and behavioural questions: a forward-looking predictive design that links readiness to subsequent project transitions, a matched-pair analysis of structurally similar economies with divergent CBDC trajectories, and the integration of micro-level adoption and survey evidence that cross-country structural indicators cannot capture.

Several items previously listed here are completed in the present revision and are reported in Sect. 4.6 and the Robustness Results Appendix, namely external validation against observed CBDC project stages, the population-threshold and size-adjusted robustness exercise, the parallel arithmetic aggregation and parsimonious institutional specification, and the theory-weighted variant of the index. The agenda that remains open is therefore narrower and concentrates on dynamic and behavioural questions. First, a forward-looking predictive design that links readiness to subsequent transitions in project stage, with explicit

treatment of reverse causality and endogenous political timing. Second, a temporal-dynamics analysis using sigma and beta convergence diagnostics on the per-year indices between 2018 and 2024. Third, a formal matched-pair toolkit application using nearest-neighbour or propensity-score matching. Fourth, a year-by-year characterisation of CRITIC weight stability with bootstrap intervals on each within-pillar weight. Fifth, extension of the pillar set as the CBDC technology and governance environment evolves. Sixth, integration of the index with micro-level adoption surveys to connect structural capacity with behavioural acceptance.

6 Conclusion

This paper develops and validates a multi-pillar CBDC Readiness Index that measures the structural capacity of countries to introduce central bank digital currency safely and at scale. The central finding is that CBDC preparedness is not primarily a question of who launches first, but of which jurisdictions possess the digital, financial, institutional, and macroeconomic foundations necessary to sustain implementation without destabilizing monetary transmission or banking intermediation.

The global distribution of readiness is highly uneven and right-skewed: a narrow group of digitally advanced, financially deep and institutionally robust economies rank at the top, while a broad set occupies intermediate bands with deficits in one or more pillars. Robustness analyses show leader and laggard classifications are stable across weighting schemes, aggregation rules, coverage thresholds, leave-one-out exercises and resampling, so the patterns are structural rather than methodological, while intermediate ranks remain appropriately sensitive. The external validation confirms that structural readiness is related to but distinct from project-stage advancement and from economic development.

For policymakers, the index functions as a dashboard rather than a ranking device. Decomposing readiness into five pillars lets authorities identify binding constraints and sequence reforms: weak digital infrastructure points to connectivity and cybersecurity investment, inclusion gaps to expanding access before retail issuance, institutional deficits to data governance and legal clarity, and macro vulnerabilities to stabilisation before scaling. For researchers, it provides an auditable structural measure that clarifies why identical announcements produce heterogeneous effects.

Several limitations define future directions. The index does not measure behavioural trust, political legitimacy, dynamic banking adaptation or specific design quality; future work can integrate micro-level data, extend to regional sub-indices, and adapt pillars as CBDC technologies evolve.

The overarching conclusion is straightforward: CBDC outcomes depend at least as much on slow-moving structural fundamentals as on the timing of individual launches, so sound policy requires an explicit assessment of digital, financial, institutional and macro conditions before ambitious implementation. The index provides a rigorous starting point for that evaluation.

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Declarations

Conflict of interest The authors declare that they have no relevant financial or non-financial interests to disclose.

Ethical approval This study relied exclusively on secondary data obtained from publicly available sources. No human participants, primary surveys, experiments, or confidential personal data were involved. Ethical approval was therefore not required.

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